

WORKSHEET

With Worksheet, you can easily perform complex calculations for budgeting, forecasting, statistical analysis, engineering—whatever uses you can imagine for a spreadsheet. The program lets you customize your spreadsheet to meet your specific needs.

Worksheet functions enable you to find data quickly, copy and delete files, and print all or part of a spreadsheet. Using the Merge function, you can save selected data from your spreadsheet in either XENIX or MS-DOS format.

This chapter tells you how to create and open Worksheet files and how to use the functions to prepare and print a spreadsheet.

Using Worksheet

To create a spreadsheet, open a new file from the Main Menu. Begin by positioning the marker on `Worksheet` in the application line. Press `[ENTER]` to load the application without specifying a file. Worksheet displays a prompt at the bottom of the screen:

Enter filename:

Type a new filename, and press `[ENTER]`. If you do not type the extension to the filename, `.wks`, the application assigns it automatically.

Note: You can open a worksheet file without specifying a filename by merely pressing `[ENTER]` at the prompt. However, when you exit the file, the application does not save its contents. To avoid losing data in such a case, you can merge the unnamed file to another file (whose name—new or existing—you specify).

The names of existing worksheet files appear on the Main Menu in the column below `Worksheet`. To open an existing file, move the marker to the application name. Then, indicate the name of the file in either of the following ways:

- Press `[ENTER]` to load the application, and then enter the filename when the prompt, `Enter filename:`, appears at the bottom of your screen.

Worksheet: Worksheet file

11/20/85 10:30am

<ESC> = Commands

	1	2	3	4	5	6	7
1							
2							
3							
4							
5							
6							
7							
8							
9							
10							
11							
12							
13							
14							
15							
16							
17							
18							
19							
20							
Select command							
R 1 C 1							

If you use the standard column width (10 characters), a window shows 7 columns and 20 rows. You can have as many as 99 rows and 99 columns in each worksheet, provided that your setup offers sufficient memory. In the following list, the circled numbers correspond to the positions shown in Figure 5.1.

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- ② Column numbers appear across the top of the screen in the column label area.
- ③ Each intersection of a row and a column is a *cell*.
- ④ The highlighted rectangle is the *entry marker*. Use the entry marker to select a cell for data entry.
- ⑤ The data entry area displays the data you enter in the marked cell.
- ⑥ The command line either prompts you to select an activity or displays the activity in which you are currently engaged. Error messages and warnings also appear on this line.
- ⑦ The cell status line indicates the cell currently highlighted by the entry marker (R 1 C 1). Text data contained in that cell appears to the right of the cell number. If the current cell contains numeric data, the cell status line specifies only the word *number*.

To enter data on the worksheet, position the entry marker on a cell, and type the data on the data entry line. The cell displays the data at the same time. If the first character is numeric (– or +, 0 through 9, or .), the command line changes to read, *Enter number*. Otherwise, the command line prompts, *Enter text*.

Press **ENTER** or one of the arrow keys to store the contents of the entry line in the cell. If, in a numeric entry, there are too many characters to the left of the decimal to display using the format for the cell, Worksheet retains the entire number, but displays a series of asterisks at the cell location. Undisplayable characters to the right of the decimal remain in memory, but the program rounds the number displayed in the cell to approximate the complete value.

The Help Screens

The Worksheet help screens contain brief summaries of the functions and ways to use them. When you need help, press **ESC** and then **?**. The screen most appropriate to your current situation on the Worksheet is displayed. Press **CTRL** **C** to return to the Worksheet.

If there are multiple screens that correspond to your current situation, press **ENTER** to display the next screen. Pressing **P** returns you to the previous screen.

Marker Movement

Use the marker movement keys to move the entry marker from cell to cell and to the row and column labels. The arrow keys move the marker a cell at a time, in the direction of the arrow. Press the other marker movement keys or key sequences to move the marker more rapidly. See Figure 5.2.

Marker Movement

Key	Moves the Marker
	1 cell (or column label) to the right
	1 cell (or column label) to the left
	1 cell (or row label) up
	1 cell (or row label) down
 	To the last column on the screen, from the last column to the last column on the next screen
 	To the first column on the screen, from the first column to the first column on the previous screen; to the row label if the entry marker is in Column 1
 	To the first row on the screen, from the first row to the first row on the previous screen; to the column label if the entry marker is in Row 1
 	To the last row on the screen, from the last row to the last row on the next screen
 	To Row 1 or to the column label if the marker is in Row 1

Figure 5.2

The Worksheet Functions

The Worksheet functions appear at the bottom of the screen when you press . The numbered functions on the first line are specific to Worksheet. The second command line displays general DeskMate commands, with the addition of Calc, which is unique to Worksheet.

1-Find 2-Text 3-Formula 4-Frmat 5-Buffer 6-Merge 7-Select 8-Copy 9-Del 0-Insert
 Calc  Help  Print  Cancel  Quit  Desk accessories

To select a Worksheet function, press the appropriate number key (after pressing **[ESC]** to display the command lines). You can use function keys (**[F1]**, **[F2]**, **[F3]**, etc.) to select the functions if your terminal provides them and your termcap file supports them. For example, you can select the Find function by pressing **[ESC]** and then **[1]**, **[ESC]** and then **[F1]**, or **[F1]** by itself.

Find

Press **[ESC]**, then **[1]** to search for an alphanumeric string or a specific cell. The following prompt appears on the command line:

Specify string or cell

When searching for a character string, the function searches any cell that contains text. The search begins at the current marker position and covers all cells that come after it (from left to right to R99 C99). It ignores numeric entries and blocks of text entered using the Text function.

A cell containing characters other than the search string is nevertheless a match. The search string **day**, for instance, matches any cell after the current marker position that contains the name of a day of the week: Sunday, Monday, etc.

Enter the cell contents that you want to find. The function finds the first occurrence of the string and displays the cell containing it. The window shifts so that the cell you specified appears highlighted in the upper left corner of the screen.

If you type no characters between uses of the function, it does not prompt you each time for a new search string. Instead, the function executes the search immediately after you press **[ESC]** and then **[1]**. The screen displays the next occurrence of the search string you entered already.

When searching for a specific cell, you must enter a valid cell number. Type **R**, followed by the row number, then **C**, and the column number. For example, type **R2C3** **[ENTER]** to find the cell located on Row 2 of Column 3. The entry marker moves to and highlights the cell you specified.

If you specify only a row number, the marker moves to the specified row in the current column. In a like manner, the marker moves to a specified column number in the current row.

Text

If you require more space for text than a single cell, or if you want the worksheet to appear formal, you can select a block of cells in which to enter text. The perimeter of the selected block defines the text boundaries, rather than each individual cell in the block. In effect, this function creates a window for writing, editing, and manipulating text.

To create a text area in your worksheet, select a block of cells using the Select function (described later in this chapter), and press [ESC], then [2]. To edit an existing text block, position the marker anywhere in the block and press [ESC] followed by [2].

As you type, words wrap to the next line. If you type a character over existing text, the text shifts to the right (insert mode). [BACKSPACE] deletes the character preceding the cursor and [CTRL] [X] deletes the character at the cursor.

To exit the Text function and save the Text block as it currently appears, press [ESC] and then [Q]. To restore text to pre-edited condition, or to cancel a new block selection for text, press [CTRL] [C] (or [BREAK]).

Within limits, you can reformat existing text blocks into larger or differently shaped blocks in your worksheet. To do so, select a block of cells (using the Select function) including at least 1 cell of the existing text block that you wish to reformat. Press [ESC] and then [2]. The text block is reformatted into the newly selected block. Edit the block or press [ESC], then [Q] to exit the Text function.

When you delete rows or columns or insert them in your worksheet, the text block shifts position along with its surroundings.

Formula

To enter a formula into a cell, position the entry marker on that cell. Press [ESC] followed by [3] to mark the cell for formula entry. Then, enter the formula on the data entry line.

To enter a formula into several cells, select the cells using the Select function. Next, mark them for formula entry using [ESC] and then [3], and enter the formula.

As the examples on the following pages show, formulas in Worksheet can refer to cell locations as *operands* (numbers on which an operation is performed). But while you enter a formula by referring to specific cells, the references are variable; they change when the formula changes location. This feature enables you to insert or delete rows and columns without having to adjust the formulas yourself; the formulas still compute the values you intended them to compute.

Consider a formula that finds the sum of Rows 3, 4, and 5 (in any column). The formula, located in Row 6, is `SUM(R3)`. Delete Row 1, and the formula adjusts to add the values in Rows 2, 3, and 4. The formula, now in Row 5, reads `SUM(R2)` after the deletion.

Now add 3 rows between Row 1 and Row 2, and the formula adjusts again to find the sum of the values in Rows 5, 6, and 7. The formula, now located in Row 8, has become `SUM(R5)`.

Finally, add a row between Rows 5 and 6, within the series of values totaled by the formula. The formula adjusts this time to include the new row, finding the sum of the values in Rows 5, 6, 7, and 8. In each instance, the formula accommodates the changes so that it still applies to the cells for which you intend it.

Rather than typing each cell location in a formula, you can display a cell location on your entry line by positioning the marker in that cell. After you choose the Formula function, begin typing the formula itself. At any point that you need to type a cell location, you can move the cursor to the cell and displays that cell location on the entry line. Then, continue to type your formula.

Repeat the process to specify other cell locations elsewhere in the formula. Press `[ENTER]` when you finish typing the formula, or press `[CTRL] [C]` to cancel the formula entry.

You might, at some point, want to refer to a fixed cell location in a formula. To prevent the reference from being adjusted, precede it with `#` in the formula. The `#` indicates a constant, rather than variable, cell location. For example, `#R2C11` specifies a fixed cell location, no matter where you relocate the formula.

Another special character lets you enter a value in a cell at the time of calculation (using the Calc function). Enter `?` as the first character in a formula cell to prompt for a constant used in a calculation.

Within a formula, parenthetical operations are performed first, multiplication and division second, and addition and subtraction last. When parentheses are nested, the innermost operations are performed first. If parentheses or the type of operation do not determine the order of computation, the operations occur from left to right. Figure 5.3 shows the operators available for use in formulas.

To erase values in formula cells while leaving the formulas intact, press **CTRL** **F**.

To edit an existing formula, position the marker on the formula cell, and press **ESC** followed by **3**. Left and right arrows enable you to position the cursor within the formula. Delete characters using **CTRL** **X** or **BACKSPACE**. Any characters you type are inserted at the cursor. When you finish revising a formula, press **ENTER** to save the revisions or **CTRL** **X** to cancel them.

Formula Operations between 2 cells:

+	Addition	*	Multiplication
-	Subtraction	/	Division
		!	Power

for a row, column, or selected block:

AVG	Average (mean) of the values	MIN	Minimum value
		RMT	Cumulative sum of the row
CMT	Cumulative sum of the column	SUM	Sum of the values
MAX	Maximum value		

for a cell:

ABS	Absolute Value	LOG	Logarithm
ATN	Arctangent	SGN	Sign
COS	Cosine	SIN	Sine
EXP	Exponential	SQR	Square root
INT	Integer truncation	TAN	Tangent

Figure 5.3

The following are examples of formulas using some of these operations. The examples demonstrate accepted syntax with the various operators. The program recognizes both lower- and uppercase characters.

? CONST A Sets up a cell to receive a constant (labeled CONST A) to be entered at the time of calculation.

(R1 + R2) Adds 2 cells in the same column.

(C3)-(R1C4) Subtracts the value of the cell in Row 1, Column 4 from the value of the cell in Column 3 of the current row.

C1*C2 Multiplies two cells in the same row.

(R1C2)/100 Divides the value of the cell in Row 1, Column 2 by 100.

C6R3!3 Calculates the cube of the value of the cell in Row 3, Column 6.

ABS(R2C3) Multiplies the value of the cell in Row 2, Column 3 by -1 if (and only if) it is a negative number (absolute value).

ATN(R3C5) Displays the arctangent of the value in Cell R3C5—the angle (measured in radians) that has a tangent equal to the value in Cell R3C5. (Use $\text{ATN(R3C5)*57.27273}$ to display the arctangent in degrees.)

AVG(C1) Adds all the values in the current row, beginning with Column 1, over to the current cell, skipping any non-numeric data, and then divides by the number of numeric cells added.

CMT(#R4C3) Gives a cumulative total for Column 3, beginning with Row 4. For example:

	[3]	[4]
[4]	5.00	
[5]	10.00	
[6]	20.00	

If you select Cells 4, 5, and 6 of Column 4 and enter the above formula, calculation produces the following results:

	[3]	[4]
[4]	5.00	5.00
[5]	10.00	15.00
[6]	20.00	35.00

COS(R3C5) Displays the cosine of the value in Cell R3C5. (Use COS((R3C5)*.01745329) if the value in Cell R3C5 is in degrees instead of radians.)

EXP(R4C3) Displays e raised to the power of the value in Cell R4C3 (Napierian, or natural exponential e^x).

INT(R4C2) Displays the truncated value of Cell R4C2.

LOG(R2C4) Displays the logarithm to the base 10 of the value in Cell R2C4.

MAX(C4) Displays the maximum value in the current row, beginning with Column 4, over to the current cell, skipping any non-numeric data.

MIN(R1) Displays the minimum value in the current column, beginning with Row 1, down to the current cell, skipping any non-numeric data.

RMT(#R4C3) Gives a cumulative total for Row 4, beginning with Column 3. For example:

	[3]	[4]	[5]
[4]	5.00	10.00	20.00
[5]			

If you select Cells 3, 4, and 5 of Row 5 and enter the above formula, calculation produces the following results:

	[3]	[4]	[5]
[4]	5.00	10.00	20.00
[5]	5.00	15.00	35.00

SGN(R3C4) Displays the sign of the value in Cell R3C4 (displays 1.00 if the argument is positive or zero, and -1.00 if it is negative).

SIN(R2C1) Displays the sine of the value in Cell R2C3. Use SIN((R2C1)*.01745329) if the value in Cell R2C3 is in degrees instead of radians.

SQR(R5C1) Displays the square root of the value in Cell R5C1.

SUM(R5) Displays the sum of the values in the current column, beginning with Row 5, down to the current cell, skipping any non-numeric data.

TAN(R2C3) Displays the tangent of the value in Cell R2C3. Use TAN((R2C3)*.01745329) if the value in Cell R2C3 is in degrees instead of radians.

Format

Use the Format function to change column widths or data format. You can change individual column widths or change all column widths at once. Similarly, you can change the format for data in a single cell or for a selected block of cells.

Initially, a spreadsheet displays 10 characters per column. To change column width, press **[ESC]**, then **[↑]** while the entry marker is in Row 1. The marker moves up into the column label

area. Then, if you want to change the width of a particular column, move the marker to the column you want to change.

Next, press **[ESC]** and then **[4]**. The following message appears on the command line:

Specify All,width or width

If you want to change the width of all columns, type **All**, followed by the new width; and then press **[ENTER]**. For example, change the width of all columns to 15 by typing:

All,15 [ENTER]

Change only the width of the current column by entering the width you prefer and omitting **All**. To change the current column to 15 characters, for example, type the following:

15 [ENTER]

To use the Format function to specify a certain format for existing cell data, place the entry marker on the cell you want to change or select a group of cells (using the Select function). Press **[ESC]** and then **[4]**. The screen shows:

Specify format types (L, R, D, I, \$)

At the data entry line, enter the letter or letters for the format you wish to use, as follows:

L = left-justified

R = right-justified

D = decimal (up to 7 decimal places)

I = integer (whole number)

\$ = dollar format (2 digits after the decimal)

Notice that you can specify more than one format modifier. You can combine a justification format (L or R) with an indication of precision (D, I, or \$). For example, **LI [ENTER]** indicates a left-justified, integer format.

If you do not customize cell formats, text is routinely left-justified, while numeric data is right-justified in dollar format.

When you specify decimal (D) format, a second prompt enables you to specify the number of decimal places to display. For practical purposes, do not specify more than 7 decimal places. You can store numbers on your worksheet with more than 7 digits after the decimal, but precision is lost after 7.

Buffer

The Buffer function lets you duplicate a portion of your worksheet in the copy buffer. Then, you can insert the contents of the copy buffer in another area of the worksheet.

Begin by selecting a block using the Select function. (An upcoming section explains the use of the Select function.) Press **[ESC]** and then **[5]** to place the selected block of your worksheet in the copy buffer. If the selected area includes a partial text block, a message appears:

Partial text areas in source, continue? (Y/N)

Press **[Y]** to copy the selected cells even though they do not include the complete text block. Press **[N]** to cancel the attempt.

Use the Insert function to place the contents of the copy buffer in another area on the worksheet.

Merge

Press **[ESC]**, then **[6]** to merge data from one file into another. If you begin by selecting a block of data, the function lets you save the block to a file that you specify. You can indicate either XENIX or MS-DOS® format for the file. If you do not select a block of data before invoking the Merge function, the screen prompts you for the worksheet file that you want to insert at the current marker position.

If you begin by selecting a block of cells, the screen prompts:

XENIX or MS-DOS format? (X/M): X

Press **[ENTER]** or **[X]** to save the selected cells in a XENIX file. Or type **[M]** to indicate that the file is to have MS-DOS format. A user with an MS-DOS version of DeskMate can load a file of that type from Worksheet.

After you determine the format of the destination file, a second prompt appears:

SAVE Enter filename:

At the data entry line, enter the name of the file in which you wish to save the selected block. If you select a block that includes part of a text block, a prompt appears on the command line to tell you that you are merging a partial block of text. Press **[Y]** to execute the command and merge the partial text block, or press any other key to cancel.

If you do not select a block of cells, this alternate prompt is displayed:

LOAD Enter filename:

Type the name of the worksheet file that you want to load and insert at the current marker position. Press **[ENTER]** to execute the Merge command. The Merge command inserts the file you specify, overwriting any cells in the current worksheet that overlap the inserted file.

If the file you want to load would write data in cells occupied by a text block or exceed the capacity of the current file, Worksheet cancels the Merge command and identifies the problem on the command line. Press any key to continue using your file.

Select

The Select function lets you define a block of cells on the worksheet. After you select a block of cells, you can use other commands (Formula, Text, Format, Buffer, Merge, Copy, Delete, Print) in such a way that the command applies to the entire block of cells.

Position the entry marker on the first cell you want to include in the block, and press **[ESC]** and then **[7]**. Using the arrow keys to increase or decrease the selected area, move the marker until it covers the entire area that you wish to include. All selected cells are highlighted.

After you select the block, execute another Worksheet function or print the selected block, as your require. If you choose not to use the selected block, pressing **[ESC]** and then **[7]** again cancels the selection mode.

Copy

The Copy function enables you to duplicate the contents of your worksheet in a Text (ASCII) file. You can copy the entire file or a selected block of cells. Use the Select function if you prefer to define a portion of the worksheet to copy.

Press **[ESC]** and then **[8]** to copy the file or selected block. The following prompt appears:

Enter copy filename:

Enter a new or existing document filename, including the extension, .doc. For any file not in the current directory, type the pathname. If you specify an existing file, remember that the Copy function overwrites its contents.

Delete

The Delete function enables you to erase the contents of a cell or block of cells or to eliminate an entire row or column. To erase data, position the marker on a single cell or define a block of cells using the Select function. Then, press **[ESC]**, followed by **[9]** to delete the data in the highlighted area.

To delete an entire row or column, position the marker at the top of a column or at the left end of a row you want to delete. Use **[ESC]** and then **[←]** or **[↑]** to move the marker to the row or column label. Then, press **[ESC]**, followed by **[9]** to delete the entire row or column.

Data entries from succeeding rows or columns shift to fill the void caused by deletion of an entire row or column. All formulas adjust to reflect the deletion.

Text blocks remain intact, shifting upward and to the left, if necessary, to accommodate deletion of a row or column. If a text block shifts in such a way that it “overlaps” existing data, you lose the data in the overwritten cells. Before Worksheet allows that to happen, it alerts you to the conflict and asks you to confirm the deletion by pressing **[Y]**. Press any other key to cancel the deletion.

Insert

Use the Insert function to insert the contents of the copy buffer at the current location of the entry marker or to insert a blank row or column into your worksheet.

On the worksheet, press **[ESC]** and then **[I]** to insert the contents of the copy buffer at the current entry marker position. You can insert a row of data only into another row; a column of data, only into another column.

If the destination area overlaps a text block or if the contents of the copy buffer will not fit, Worksheet cancels the command. Press any key to continue using the file.

To insert a blank row or column, position the marker on the row or column at which the insertion is to occur. With the marker at left end of the row or the top of the column, press the **[ESC]**, then **[←]** or **[↑]** to move the marker to the row or column label. Then, press **[ESC]** and then **[I]** to insert a blank row or column.

If the insertion causes existing cells to fall beyond Row or Column 99, Worksheet deletes those cells. Before it does so, however, a message alerts you to the deletion of data that would result. Press **[Y]** to confirm the insertion; press any other key to cancel it.

All formulas and text blocks adjust to reflect the row or column insertion. Text blocks shift downward and to the right, if necessary, to accommodate insertions.

Calc

The Calculate function performs the calculations you indicate in the formula cells of your worksheet.

Press **[ESC]** and then **[=]** to calculate the worksheet you set up or to recalculate whenever you make changes. The Calculate function computes data according to cell contents, from left to right and top to bottom. Any non-numeric data is skipped.

If a calculation requires values for constants, a prompt appears on the command line each time the formula requires a value. Enter the requested value to proceed with the computation.

Printing Data

Press **[ESC]**, then **[P]** to print the entire worksheet or a limited area of the worksheet defined using the Select function. A series of printer settings appears before the system begins to print. Check the default settings on the screen; make changes or simply press **[ENTER]** to accept each value.

When all settings are satisfactory, press **[ESC]** and then **[Q]** to begin printing.

Exiting Worksheet

To exit Worksheet, press **[ESC]** and then **[Q]**, which returns you to the previous level of operation and saves any changes you made to the current file. Press **[ESC]** and then **[X]** to cancel newly entered changes. A prompt appears on the command line:

Cancel? (Y/N)

Press **[Y]** to confirm the command, or press **[N]** to keep the file open.

FILER

The Filer application provides information storage and retrieval. Filer is easy to set up and maintain. Its flexibility permits you to set up a custom data entry form for each file. The form you devise determines the specific information you want for each record and the format for the information. You can use any of the information on the form for sorting or locating your records.

This chapter describes the procedures for using Filer and its functions to manage your records and make printed copies of them. You need to be familiar with certain terms that appear frequently in the chapter.

A *form* is a series of questions or labels that indicates what information you want to store and a corresponding series of blanks in which you can enter the requested information.

Each blank to be filled on the form is a *field*, and when you enter data in the fields on a form, you create a *record*. Each file that you construct using Filer consists of a series of records created in this manner.

Using Filer

To create a data file, open a new file from the Main Menu. Begin by positioning the marker on **Filer** in the application line. Press **ENTER** to load the application without specifying a file. Filer displays a prompt at the bottom of the screen:

Enter filename:

Type a new filename, and press **ENTER**. If you do not type the extension to the filename, **.fil**, the application assigns it automatically.

Note: You can open a file in Filer without specifying a filename by merely pressing **ENTER** at the prompt. However, when you exit the file, the application does not save its contents. To avoid losing data in such a case, you can merge the unnamed file to another file (whose name you specify).

After you enter a new filename, the following prompt appears:

File doesn't exist, Create ? Y or N?

Press **[Y]** to create the new file, or press **[N]** to enter a different filename. Press **[CTRL] [C]** to cancel and return to the Main Menu.

The names of existing Filer files appear on the Main Menu in the column below **Filer**. To open an existing file, move the marker to the application name. Then, indicate the name of the file in either of the following ways:

- Press **[ENTER]** to load the application, and then enter the filename when the prompt, **Enter filename:**, appears at the bottom of your screen.
- Use the arrow keys to position the marker on the filename, and press **[ENTER]**. (If you have a long list of files, you may discover that the **Find** function saves you time. The **Find** function is described at a later point in this chapter.)

The first record of the file appears.

There are 5 different screens in the Filer application: the **Form** screen, the **Add** screen, the main **Filer** screen, the **Find** screen, and the **Display** screen.

When you create a file, you move immediately to a blank **Form** screen. Use it to arrange the labels and fields that define the contents of the file. It is this form that you use for collecting and presenting data for each record. The section devoted to the **Form** function, later in this chapter, includes directions for using the **Form** screen.

To begin entering records, proceed to the **Add** screen. The **Add** screen displays your form so that you can type data in response to the labels. It redisplayes the blank form repeatedly to enable you to add a series of records.

When you are ready to review and manipulate your records, move to the main **Filer** screen. It is this main **Filer** screen that appears when you open an existing file (one for which you have already defined a form). It displays the records you created so that you can read and revise them. It also gives you access to the all the other screens so that you can revise the form, add records, or devise reports.

The Find screen enables you to locate a certain record or group of records. It presents the form you devised and lets you enter search criteria in any of the fields you choose. It also offers a means of setting up reports to print or display.

Finally, the Display screen lets you see selected data in a horizontal, report format. The choices you make on the Find screen determine the composition of the displayed report. Instructions for using these 2 screens appear in the sections devoted to the Find function and the Display function.

The Help Screens

The Filer help screens contain brief summaries of the functions and ways to use them. When you press **[ESC]** and then **[?]** to select the Help command, a help screen appears. The particular screen that you see depends upon the type of activity in which you are engaged.

If multiple help screens are available in a certain situation, pressing **[ENTER]** displays the next screen. Press **[CTRL] [C]** to return to the screen from which you sought help.

Marker Movement

Use the arrow keys to move the marker a character or line at a time. Press the other designated marker movement keys or key sequences to move the marker more rapidly. See Figure 6.1.

Marker Movement

Key	Moves the Marker
	1 character to the right
	1 character to the left
	To the first character in the label or data area directly above
	To the first character in the label or data area directly below
 	To the beginning of the next record on main Filer screen
 	To the beginning of the previous record on main Filer screen
 	To the first Find match or to the first record on file
 	To the last Find match or the last record on file
	Back and forth between the label and data areas on the Form screen
 	To the first Find match or to the first record on file

Figure 6.1

The Filer Functions

The Filer functions help you maintain your records. To display the functions available on the main Filer screen, press . The command lines shown below appear at the bottom of the screen. To select a Filer function or a general DeskMate command, press the corresponding key.

```

1-Find  3-Display  4-Print  5-Form  6-Merge  9-Delete  0-Add
<?> Help  <P> Print  <X> Cancel  <Q> Quit  <=> Desk Accessories

```

Some of the functions (Find, Form, Add, and Display) offer their own command menus, different from those shown above. This section explains the use of each Filer function and of the commands each, in turn, offers.

Find

Press **[ESC]** and then **[1]** to search for and find a record or group of records. The Find screen displays a blank form. Press **[ESC]** to display the special Find functions—Equal, Greater, Less, Reset, Mark—at the bottom of the screen.

For each field, type the *search data*, consisting of a series of characters that the Find function uses to select records. To skip a field, press **[ENTER]** or **[↓]**.

You can include wildcard characters in your search data if you use Equal (=) as the Find operator. Use * and ? in wildcards as matching characters, described in “Wildcards” in Chapter 1.

The different Find operators enable you to specify the relationship to exist between the search data in a field and the data that the Find function selects. The possible relationships are Equal to, Greater than or equal to, and Less than or equal to. In alphanumeric fields, the operators pertain to alphabetical order.

To set the Find operator, press **[ESC]**, and then press the number key that corresponds to the operator you want. If you do not specify an operator, Filer assumes that the operator is Equal. Choose the function any time the marker is over the appropriate field—before, while, or after you enter the search data.

Press **[ESC]** and then **[Q]** to begin the search. If search criteria exist for more than 1 field, Find searches for records that match all the criteria. The function finds all records that match the criteria, and displays the first one. Use **[ESC]** with **[→]** or **[←]** to scroll forward or backward through the records.

Reset. Press **[ESC]**, then **[5]** to eliminate the Find criteria. The function erases any search data, so that all your records are available to you again. Enter new search data, or press **[ESC]** and then **[Q]** to return to the first record on the main Filer screen.

Mark. Press **[ESC]** and then **[7]** to “mark” or “unmark” fields for use with the Display and Print functions. Marked fields are visible—displayable or printable. An asterisk appears in the label area of all visible fields. Invisible fields do not appear on the Display screen or printed reports. When you first create a form, all fields are marked.

Display

Press **[ESC]**, then **[3]** from the main Filer screen to display a report listing all records that match the Find criteria.

The report consists of a label area across the top of the screen and a record area occupying the rest of the screen. The label line identifies each of the fields marked for display on the Find screen. The labels are arranged horizontally and spaced to correspond to the columns of data in the record area.

The record area displays the data in vertical columns under the appropriate field label. The length of each field is determined by the longest item of data entered for that field. Two spaces are inserted between each field.

If the records take up more than 80 characters across the screen, the records (label line and data lines) wrap to the next line, indented 5 spaces. If more records meet the selection criteria than the screen can contain, use the marker movement keys to scroll through the records. (See “Marker Movement.”)

If you do not wish to display certain fields, unmark those fields on the Find screen.

If you press **[ESC]** while displaying a set of records in the horizontal format, the bottom of the screen displays special functions that you can use from the Display screen.

Print. Press **[ESC]** and then **[4]** to print the entire report currently displayed on the screen. The report is printed in the same format (horizontal) that the screen shows. Examine the default values on the printer setting screen, make changes as needed, and press **[ESC]** and then **[Q]** to begin printing.

Use this special Print function in Display only when you want to print the entire report. Use **[ESC]** and then **[P]** when you want to print selected text.

Select. Press **[ESC]** and then **[7]** to use the Select function to define a rectangular block of text. Use the marker movement keys to increase or decrease the highlighted area. When the dimensions are satisfactory, press **[ESC]**, then **[8]** to copy the selected data to a Text (ASCII) file, or press **[ESC]** and then **[P]** to print.

Copy. Press **[ESC]**, then **[8]** to copy a block of data (defined using the Select function) to a Text (ASCII) file. The Copy function prompts you for a filename at the bottom of the screen. Type a suitable filename, and press **[ENTER]**.

Press **[ESC]** and then **[Q]** to return to the main Filer screen when you are finished with the Display function.

Print

Press **[ESC]** and then **[4]** at the main Filer screen to print all records that match the Find criteria. When you select this function, the printer settings appear. Check the default settings on the screen; make changes, or simply press **[ENTER]** to accept the displayed values. When the displayed values meet your needs, press **[ESC]** and then **[Q]** to begin printing.

The report is printed in a horizontal format similar to that displayed using the Display function. A label line across the top of the page displays headings for each column of data. The data for your chosen records extends in columns below each heading. Each field appears under its corresponding label. The longest item of data entered for that field determines the length of each field. Two spaces separate each column.

To print blank forms, individual records, or a selected block of the Display screen, see “Printing Records” in this chapter.

If the printed record requires more space than is available on 1 line, the records (label line and data lines) wrap to the next line, indented 5 spaces. If you do not wish to print certain fields, unmark those fields in Find.

Form

The Form screen appears whenever you create a new Filer file. Initially, the screen displays no labels or fields. Follow the directions in this section to define them for your recordkeeping form. Once you create the form, you can begin to add records.

The Filer program lets you make changes to your form after you have begun to add records. Using the Form function, you can return to the Form screen by pressing **[ESC]** and then **[5]** from the main Filer screen.

Using various functions on the Form screen, you can change labels or fields or revise the sort order for records. For each item you want to include in your records, you must set up a label and

a field area on the Form screen. You can define a form with a maximum of 21 fields.

Defining a Label. The left side of the screen contains *field labels*, tags that identify the data in the field or remind you what data to enter. A label always begins at the left margin.

When creating a new form, you can begin immediately to type your first label. On an existing form, you can add label on a blank line or insert a line for a new label using the Insert function, **[ESC]** and then **[0]**.

Type 1 or more characters (a maximum of 15) on a line in this area to create a new label on that line. Press **[ENTER]** after you complete the label. The unused label area fills with a line of dots and a colon. The marker shifts to the field area on the same line.

Note: To leave blank lines between field labels, press **[ENTER]** or **[↓]** before you enter any characters in the label area. On an existing form, use the Insert function to insert blank lines between labels.

Defining a Field Area. Press **[ENTER]** in the first position of the field area to display a line 59 spaces long. This represents the maximum number of alphanumeric characters that you can enter in a 1-line field. (See “Number” if you want to set up a numeric field.) Press **[ENTER]** again to move to the label area of the next line.

You can define a field to be shorter than 59 characters if you wish. Instead of pressing **[ENTER]** immediately when the marker shifts to the field area (the right side of the screen), define the field area one space at a time. Press **[ESC]** and then **[0]** once for each space you want to devote to the data field.

You can extend a field beyond 59 characters to include multiple lines—a maximum of 255 characters. Press **[ENTER]** at the beginning of a line to fill the line with blanks. Press **[↓]** and then **[ENTER]** again to fill the next line with blanks. Continue in this way until you define the maximum field space (4 complete lines and part of a fifth). The label area for a multi-line field is blank on each line below the first one.

Besides modifying the length of each field, you can add *edit characters* to the field. Edit characters indicate a certain type of required entry—in either alphanumeric or numeric data fields. A common use of edit characters is the formatting of telephone numbers, for example:

(...) ...-....

The parentheses, the space, and the dash are edit characters that indicate the placement of the area code and local dialing sequence; the cursor skips over them during data entry.

Special Form functions appear at the bottom of the screen when you press **[ESC]**:

	1-Order	3-Number	9-Delete	0-Add
<?> Help	<P> Print	<X> Cancel	<Q> Quit	<=> Desk Accessories

Order. In data entry, Filer sorts and displays your records in the order you set here. The ordered, or key, fields are sorted first, in ascending order. Then, Filer sorts the rest of the fields in their order of appearance on the screen.

If you do not establish priority numbers, Filer sorts fields consecutively from the top down. Once you do assign numbers, the priority number of key fields appears in the label area on the screen. You can assign priority numbers from 1 through 9.

Press **[ESC]** and then **[1]** to make the field under the marker a key field. The screen shows:

>Priority Number:

Enter the priority for this key field. If you press **[ENTER]** without entering a priority number, Filer assigns it the next available number (1 if no fields are ordered yet).

If you choose a field that is already a key field, press **[ENTER]** to leave the priority unchanged, or enter a new priority. If you use an existing priority, the priority numbers shift to allow for the substitution.

A priority number of 0 removes the key status of a field and reorders the key fields.

Number. Press **[ESC]**, then **[3]** to specify the field under the marker as a numeric field. A number sign (#) appears after the label (in place of the colon) to denote a numeric field. (Press **[ESC]** and then **[3]** again to change the field back to alphanumeric.)

If the field is not yet set up when you use the Number function, Filer sets it up for 12 digits, 10 to the left of the decimal point and 2 to the right. You can edit the number of digits on either side of the decimal with Add or Delete. Data entered in the field aligns around the decimal.

If you define a field before you designate it as a numeric field, the Number function right-justifies any data you enter in the field.

Delete. In the field area, press **[ESC]** and then **[9]** to delete the character at the cursor.

In the label area, press **[ESC]**, then **[9]** to delete the entire field, including the label.

In either area, pressing **[CTRL]** **[X]** deletes the character at the cursor.

Add. In the field area, pressing **[ESC]** and then **[0]** adds a character at the cursor.

In the label area, press **[ESC]** and then **[0]** (or press **[CTRL]** **[W]**) to add a new field at the current marker position. The current line and all that follow it move down to accommodate the new line.

To save a new form, press **[ESC]**, then **[Q]**. The Add screen appears so that you can begin to enter data. If you have edited an existing form, pressing **[ESC]**, then **[Q]** returns you to the first record.

Merge

Press **[ESC]** and then **[6]** to merge another Filer file into the current file. The 2 files must have identical formats.

At the bottom of the screen, a prompt appears:

Merge from:

Enter the name of the file you want to merge into the current file. If the file formats are exactly the same, copies of all the records in the (Form) file are merged into this file. If duplicate records exist in the 2 files, Filer does not merge a second copy of the identical records into the file.

Copying a Form to Use in Another File. Using the same form in 2 Filer files enables you to merge records from one to the other. Because a single character difference in format prevents merger of 2 files, make a copy of your file if you intend to reuse the form. After you set up your format (before you enter any records), use the Copy function on the Main Menu to carry out this procedure. (See “The Main Menu.”)

Delete

The Delete function lets you delete the record currently displayed. Using the marker movement keys or the Find function, display the record you want to delete. Press **[ESC]** and then **[9]** to delete the record currently displayed.

Add

The Add function displays a blank form so that you can add a new record. Enter information on the blank form, and press **[ESC]**, then **[0]** to add the new record and display another blank form. Follow this procedure to add each new record. Press **[ESC]** and then **[0]** to add the last new record and return to the main Filer screen.

On an entry screen, you always work in the overstrike mode, whether you are adding a new record or editing an existing one. In overstrike mode, each character you type replaces the character at the current marker position.

Type the appropriate data for each field, and press **[ENTER]** to move to the next field. The marker moves from field to field, starting at the top of the form. Use the marker movement keys to move through the data fields as desired.

Alphanumeric fields are left-justified. Although you can extend a field over a number of lines, words do not wrap automatically to the next line. Numeric fields, identified by #, align around the decimal or are right-justified if there is no decimal.

As you add records, they automatically sort according to the primary key field. If you used the Order function in Form, records sort by the key fields first, then by the rest of the fields in order of appearance on the form. Records sort correctly, regardless of the order in which you add them.

Printing Records

In the Filer application, there are 2 ways of printing data in your files.

When you use the Display function to produce a list in the horizontal format, you can print the **entire** list using **[ESC]** and then **[4]**, the special Print function for the Display screen.

The same function is available on the main Filer screen. Here too, it produces a printed copy of the entire file, but as a series of records rather than as a horizontally arranged report.

Use the general Print command, **[ESC]**, then **[P]**, to print records more selectively. The same command, used at the Form screen, produces a blank data entry form.

When a record is displayed on the main Filer screen, pressing **[ESC]** and then **[P]** gives you a printed copy of that record.

When you generate a list of records using the Find and Display functions, you can select a block of text to print **part** of the list. Press **[ESC]**, then **[P]** after you define a block of data using the Select function.

After you press **[ESC]** and then **[P]**, the screen lists the printer settings so that you can check the displayed values before printing. Enter any changes you want to make, or press **[ENTER]** to accept a displayed setting. When all the settings are satisfactory, be sure that the printer is properly connected and online. Then, press **[ESC]** and then **[Q]** to begin printing.

Exiting Filer

Exit your file by pressing **[ESC]** and then **[Q]** to quit the file and record all changes and additions that you made. Filer does not permit you to exit and cancel your work using **[ESC]** followed by **[X]**. The screen displays the message, **Closing** while Filer updates its memory, after which you return to the previous level of operation.

CALENDAR

The Calendar application is an event scheduler. You can use it as a general purpose calendar and planner to replace your desk calendar and date book. It records important events, their dates, and their times so that you can refer to them at any time. Beginning with the current day, Calendar provides you with a week-at-a-glance schedule.

Using the Calendar functions, you can search for an entry, print a list of events, and set the alarm for selected events. Other functions make it easy to add or delete events from your file, combine 2 Calendar files, or copy events to another file.

Using Calendar

The Calendar application relates its activity to a primary file, the default file for your home system. The application creates this file, `dmevents.cal`, in your home directory. It opens this file whenever you execute the Calendar application without specifying a filename.

To open `dmevents.cal`, your home directory need not be the current directory. Regardless of the current directory, Calendar opens the default file in your home directory. Choose the application `Calendar` on the application line, using the arrow keys to position the marker and `[ENTER]` to indicate your choice. The screen displays the Calendar and loads any data you have entered in `dmevents.cal`.

To create a new Calendar file on the Main Menu, use the Add function. Press `[ESC]` and then `[0]`, and type a name for the file, including the extension, `.cal`, at the `New file:` prompt. When the filename appears in the column under `Calendar`, the menu has created an empty file in the current directory.

The names of existing Calendar files in the current directory appear on the Main Menu in the column below `Calendar`. To open an existing file, move the marker to the application name. Then, position the marker on the filename using the arrow keys, and press `[ENTER]`.

In the upper left corner of the screen is the Weekly Time Chart. It allows you to see at a glance which blocks of time are free or committed (or overcommitted). The days of the week (Sun, Mon, Tue, and so forth) are displayed down the left side of this chart. The times of day, beginning with 12:00 a.m. (midnight), appear across the top.

Asterisks (*) mark the day and time slots for which you have scheduled an event. When event times conflict with each other, the Weekly Time Chart indicates the conflict with an exclamation point (!).

In the upper right corner of the screen is the Monthly Calendar, similar to a standard desk calendar. When you open a Calendar file, the marker rests on the current date.

In the bottom section of the screen is the Daily Events Calendar, on which you enter events. Calendar sorts and displays events in date/time order regardless of the order in which you add them to the file. If you enter more events than the Events area can display, you can scroll them onto the screen using the marker movement keys. (See “Marker Movement.”)

The remainder of this chapter explains the way to manage the data that you store in Calendar files.

The Help Screens

The Calendar help screens contain brief summaries of the functions and ways to use them. Within the Calendar application, press **[ESC]** and then **[?]** to view the help screens.

After the first help screen appears, press **[ENTER]** to see the next screen. To return to the previous screen, press **[P]**. Press **[CTRL] [C]** to return to the main Calendar screen.

Marker Movement

The marker movement keys control the date indicator in the calendar and the marker in the list of events. Pressing **[ESC]** and then **[3]** switches control from one area to the other, between Calendar and Events.

The arrow keys move the date indicator a day or a week at a time. To move the indicator a month at a time, press **[ESC]**, followed by **[→]** or **[←]**.

The event list can display 14 or 15 event lines on the screen, depending on your terminal screen size. After you enter an event on the last line, the events scroll to let you continue. To see a line after it has scrolled off the screen, press **[↑]** until the line appears again.

Use the arrow keys to move the marker a character or line at a time. Press other designated keys or key sequences to move the marker more rapidly. See Figure 6.1.

Marker Movement in Events List

Key	Moves the Marker
[→]	1 character to the right
[←]	1 character to the left
[↑]	1 line up in the current column
[↓]	1 line down in the current column
[ESC] [→]	to the beginning of the next column
[ESC] [←]	to the beginning of the previous column
[ESC] [↑]	to the first entry on the current screen or previous page
[ESC] [↓]	to the last entry on the current screen or next page
[CTRL] [T]	to the first entry in the file or to the first Find match

Figure 6.1

The Calendar Functions

The functions that are available in Calendar appear at the bottom of the screen when you press **[ESC]**. To use a function (after you press **[ESC]**), press the key that corresponds to it.

```
1-Find 2-Date 3-Calendar 5-Alarm 6-Merge 7-Select 8-Copy 9-Delete 0-Add
<?> Help <P> Print <X> Cancel <Q> Quit <=> Desk Accessories
```

Each Calendar function corresponds to a number key, shown on the first command line above. The second line reminds you of the general DeskMate functions, described earlier in this manual. This section explains the Calendar functions listed on the first line.

Find

Press **[ESC]** and then **[1]** to search for an event. An event line filled with asterisks (*) appears. If you have already used Find during the current session, the previously entered search data is displayed.

You can use the displayed settings or change any of the search data you choose. If you use the existing settings, you need only press **[ESC]** and then **[Q]** to begin the search.

To enter new search data, type the data for each field (including * and ? if you wish), and press **[ENTER]**. The wildcard character * matches any number of characters. The character ? matches any single character.

Press **[ESC]** to display the functions that are available while you are setting the search criteria. They appear along the bottom of the screen, as shown below.

```
1-Equal      2-Greater    3-Less than  4-Reset
<?> Help  <P> Print  <X> Cancel  <Q> Quit  <=> Desk Accessories
```

The Find function lets you define the relationship between the search data that you type for a field and the events you want to find. If you do not specify a relationship, the function sets Equal (=) as the Find criterion. Your options are the following:

Equal — matches only identical data.

Greater — matches any data greater than or equal to the search data.

Less than — matches any data less than or equal to the search data.

After you press **[ESC]**, press the appropriate number key to execute one of the functions listed on the bottom lines of the screen. You can choose the function anytime the marker is over the appropriate field, before, during, or after you type the data.

Press **[ESC]** and then **[Q]** to begin the search. The function finds all events that match the criteria. If more events match the criteria than the Events List can display, use the marker movement keys to scroll up or down the list of events.

To eliminate the Find criteria, press **[ESC]** and the **[1]** to redisplay the Find screen. Then, execute the Reset function by pressing **[ESC]** and then **[4]**. Each field fills with asterisks once again. Having reset the Find criteria, you can enter new criteria or press

[ESC] and then **[Q]** to exit the Find screen with the criteria reset. Calendar then redisplay the complete Events List. Exiting the file also eliminates the current find criteria.

Date

Press **[ESC]** and then **[2]** to search for and display events that fall on a certain date. At the bottom of the screen, the following prompt appears with the current date displayed after the colon.

Enter date :

Type a new date over the displayed date, and press **[ENTER]**. All events for the date specified are found, and as many as possible appear in the Events List. If more events match the date than can be displayed at one time, use the marker movement keys to scroll forward or backward through the events.

Calendar/Events

The Calendar/Events function switches the marker between the calendar (in the upper right corner of the screen) and the list of events (in the lower half of the screen). Press **[ESC]**, then **[3]** to transfer the marker back and forth.

When the command line displays `Calendar`, the marker moves to the current date on the calendar when you execute the function. With the marker in the calendar, the marker movement keys control the position of the date indicator. Moving the date indicator changes the current date.

The “current” date is not necessarily the same as “today’s” date. When you open a Calendar file, the current date is the system date; by moving the date marker, you can make any date you choose the current date. In turn, the current date determines the data shown on the accompanying Weekly Events Chart.

When the function line displays `Events`, use the function to move the marker from the calendar to the list of events. The marker shifts to the first item in the current list of events. To execute many Calendar functions (Alarm, Select, Copy, Delete), you must first move the marker to the Events List.

Movement of the marker in the Events List is described in detail in a previous section of this chapter, “Marker Movement.”

Alarm

The Alarm function enables you to set the beeper to alert you to a scheduled event. Begin by positioning the marker on an event. (If the marker is not in the list of events, press `[ESC]` and then `[3]` to switch it there using the Events function.) To define a block of events, use the Select function (described later in this chapter).

Press `[ESC]` and then `[5]` to turn on the alarm for the event or block of events that is highlighted. A message appears briefly at the bottom of the screen:

```
Merging events into Alarm file
```

If you selected a block of events, the Alarm function sets the alarm for each selected event and then turns off the Select mode.

You can set an alarm for an entry in any Calendar file. If you are working in a file other than `dmevents.cal`, however, the Alarm function merges the entry into the `dmevents.cal` file. In this way, Calendar keeps track of all alarm events in a single file.

The alarm sounds prior to the scheduled starting time for the event. You determine the amount of time between the alarm and the starting time for the event in the Events/Alarm accessory. See the discussion of Events/Alarm in “Desk Accessories” for more information.

Merge

Press `[ESC]` and then `[6]` to merge a block of events you select into another Calendar file or to merge another Calendar file into this event file.

Merging Another File into the Current File. If you do not define a block of events before selecting the Merge function, the bottom line of the screen shows:

```
Merge from :
```

Enter the name of the event file you want to merge into the current file. The merger does not alter the file you specify. The Merge function merely copies the contents of the specified file into the current file, sorting them with the events in the current file.

Merging Selected Events into Another File. If you begin by selecting events (using the Select function), the Merge function copies the selected events to another file. After you press **[ESC]**, then **[6]**, the screen shows:

Merge to :

Enter the name of the event file (new or existing) into which you wish to merge the selected events. A copy of all selected events in the current file merges into the file you specified.

Select

Use Select to define an event or a block of events. Position the marker on the first event you want to select, using the marker movement keys or the Find function. (If the marker is not visible in the list of events, press **[ESC]** and then **[3]** to switch to Events mode.) Press **[ESC]** and then **[7]**.

Use the arrow keys to place the marker over the last event you want to include in the block. As you move the marker, it highlights all selected events.

After you select the events, use one of the other commands (Merge, Copy, Delete, Alarm, or Print). If you use any other function or exit the events file before you perform one of the above operations, you cancel the selection.

Copy

The Copy function copies a previously selected block of events into a Text (ASCII) file. Begin by using the Select function to define a block of one or more events. Then, press **[ESC]**, followed by **[8]** to choose Copy. The screen shows:

To :

Enter the name of the Text (ASCII) file to which you wish to copy the events. Be sure to type the complete pathname to identify a file in another directory. The function then copies the events into the document file. If you specify an existing file, the Copy function overwrites its contents.

Delete

The Delete function enables you to eliminate an event or a block of events. Move the marker to the first event you want to delete. (If the marker is not visible in the list of events, press **[ESC]** and then **[3]** to transfer the marker from the calendar to the Events List.) If you want to define a block of events to delete, use the Select function.

Press **[ESC]** and then **[9]** to delete the marked event or the block you selected. The events are immediately deleted.

Add

Using the Add function, you can add an event to your list. Press **[ESC]**, then **[0]** to create a blank event line at the current marker position.

Press **[ENTER]** to accept the displayed date, or type over it (using the same format) and press **[ENTER]**.

Type the time at which the event begins, and press **[ENTER]**. For example, type **0745p** **[ENTER]** for 7:45 p.m. Because the function supplies zeroes for unspecified time characters and a.m. times if you do not specify otherwise, you can simply type **11** **[ENTER]** for 11:00 a.m.

Next, enter the time at which the event ends, and press **[ENTER]**.

Finally, enter a description of the event, using no more than 44 characters. After you press **[ENTER]**, the Add function inserts the new event in its proper place in the list and returns you to the current event list. Press **[ESC]** and then **[0]** to Add another event.

If you do not enter times for an event, Calendar regards it as a *day event*. Day events appear ahead of timed events in the Event List, but day events do not affect the Weekly Time Chart.

Printing Events

Press **[ESC]** and then **[P]** to print a list of the events in a file that meet the current Find criteria. If you define a block of events (using the Select function) prior to printing, **[ESC]** and then **[P]** prints only the block you selected. By using the Find (or Date) function with the Select function, you can define very specific lists of events.

Check the default settings on the printer screen before proceeding, and be sure the printer is properly connected and online. When you are ready to start printing, press **[ESC]** and then **[Q]**.

Exiting Calendar

To exit your Calendar file, press **[ESC]**, then **[Q]** or **[ESC]** and then **[X]**. Pressing **[ESC]** and then **[Q]** lets you quit work in the file and saves all changes you have made to it. DeskMate returns you to the previous level of operation—the Main Menu or your system prompt.

When you press **[ESC]**, then **[X]**, you exit the file, but you also cancel all additions and revisions you have made since you opened the file. The following prompt appears at the bottom of the screen:

Cancel? (Y/N)

Press **[Y]** to confirm that you want to cancel all changes. If you want to save the changes, press **[N]** to nullify the Cancel command.

MAIL

The Mail application handles mail that you send to and receive from other system users. After you read mail, you can delete it, file it, or forward it to another user. Or reply to the message right away—including with your reply (if you so direct) a copy of the original message.

You can write brief messages or use your text editor to compose a lengthy letter. Then, send mail to an individual or to the members of a group. Other features of the application let you distribute carbon copies of a message to selected users and verify that the message has been seen at each of its destinations.

Naturally, you can save a copy of your mail, both incoming and outgoing, in Mail files or in Text (ASCII) files.

Using Mail

The Mail distribution system is similar in its components to the system for posted mail. Mail from an individual goes to a central point—a system file that corresponds to the central post office. At that point, mail is sorted for delivery to individual users.

If there is mail waiting for you in the system file, a message appears on the left end of the status line, at the top of the screen. The word MAIL tells you that there is “new mail,” mail that you have not yet retrieved. The message appears whenever you are in DeskMate, regardless of the application you are using.

In your home directory, Mail creates a mailbox (a file) to which it routinely delivers new mail. This mailbox, your default mailbox, is comparable to your house mailbox—the one out on the street visited by your mail carrier. When you open the default mailbox, which is named `dmbox.msg`, Mail delivers any new mail to it.

You can create other mailboxes to organize the mail that you receive and copies of messages you send. These are comparable to the file folders you might set up to keep track of your correspondence, one for bills, another for insurance policies, etc.

Although `dmbox.msg`, your default mailbox, occupies your home directory, you can open it from another directory. Highlight `Mail` on the Main Menu, using the arrow keys to position the marker. Press `[ENTER]` to mark your choice. The screen displays the contents of your default mailbox, including any new mail in the system mailbox that is addressed to you.

To create a new file from the Main Menu, use the Add function. Press `[ESC]` and then `[0]`. At the `New file:` prompt, type a name for the file, including the extension, `.msg`. When the filename appears in the column under `Mail`, the menu has created an empty file in the current directory.

The names of existing Mail files in the current directory appear on the Main Menu in the column below `Mail`. To open an existing file, move the marker to the application name. Then, highlight the filename using the arrow keys, and press `[ENTER]`.

The initial screen in a Mail file lists the letters that the file already contains. Each message line in the listing includes the author's name, the date and time the message was sent, and a brief description of its contents. Use the marker movement keys to scroll through the listing if there are more messages than the screen can display (20 or 21, depending on your terminal).

To read a message, place the marker on the identification line for that message. Press `[ENTER]` to see the text of the message. Press the space bar to advance to the next screen in the message, or press `[Q]` to exit the message.

When you exit the letter, the initial screen returns, listing the contents of the mailbox.

This chapter describes the functions in the Mail accessory that let you create, distribute, and store mail. One of the functions (Option) displays a screen on which you can customize the features on your Create Mail screen. Another function displays the Create Mail screen itself. Both screens, the Option screen and the Create Mail screen, have functions of their own, described in the course of this chapter.

To return to the Main Menu, be sure the screen is displaying the list of messages, and press `[ESC]`, then `[Q]`.

The Help Screens

The Mail help screens contains brief summaries of the functions and ways to use them. Within the Mail application, press **[ESC]** and then **[?]** to see help screens that correspond to your current situation.

To see subsequent help screens, press **[ENTER]**. To display the previous help screen, press **[P]**. Press **[CTRL]** **[C]** to return directly to the Mail screen.

Marker Movement

Use the arrow keys to move the marker a character or line at a time. Pressing the other designated marker movement keys moves the marker more rapidly. See Figure 8.1.

When you create a message using the Edit function, marker movement follows the conventions of the text editor that you specify (on the Mail Options screen). DeskMate uses the editor in the Text application if you do not specify an alternate. See Figure 4.2 for marker movement in Text.

Marker Movement

Key	Moves the Marker
[→]	not used
[←]	not used
[↑]	1 line up in the current column
[↓]	1 line down in the current column
[ESC] [→]	not used
[ESC] [←]	not used
[ESC] [↑]	to the top of the screen
[ESC] [↓]	to the bottom of the screen
[CTRL] [T]	to the beginning of the file

Figure 8.1

The Mail Functions

To display the Mail functions along the bottom of the screen, press **[ESC]**. The following label line appears; it matches each of the Mail functions to a number key. Press the number key to select the corresponding function.

```
1-Find 2-Reply 3-Forwd 4-GetMail 5-Option 6-Merge 7-Select 8-Copy 9-Del 0-Create  
<?> Help <P> Print <X> Cancel <Q> Quit <=> Desk Accessories
```

To create a message, for example, choose the Create function by pressing **[ESC]** and then **[0]**. You need not wait for the label line to appear before pressing the command key to execute a function.

Find

The Find function is useful when your mailbox contains a lot of messages. Press **[ESC]**, then **[1]** to search for a particular message. The screen provides an entry line at the bottom of the screen on which you can enter search data.

Type the character string you want to find, and press **[ENTER]**. The Find function searches all messages in the file, including both the information in the headers and the content of each message. The screen displays the identification lines for any messages that contain the search data.

You can then read the messages or use other functions (Merge, Copy, Print, etc.) to manipulate the messages in the list. To return to the complete mail list for the file, press **[ESC]** and then **[Q]**.

Reply

Press **[ESC]** and then **[2]** to create a reply to the message under the marker. The Reply function displays the Create Mail screen and responds automatically in 3 of the fields.

At **To:** the Reply function enters the sender of the current message (the one to which you are replying).

At **Subject:** the function repeats the description entered for the current message.

At **Carbon Copy:** the function duplicates the list of recipients on the current message.

When you send a message created through the Reply function, Mail includes the header of the original message—the one to which you are responding.

Forward

Press **[ESC]**, then **[3]** to forward a message or group of messages to another user. To define a group of messages you want to forward, use the Select function.

When you execute the function, the following prompt appears at the bottom of the screen:

Forward to:

Type the login name of the user to whom you want to forward the message, and press **[ENTER]**. At its destination, the forwarded mail appears with a note to the recipient labelling it as a forwarded message. The word *Forwarded* appears in the heading of the message.

Forwarding a message does not eliminate it from your mailbox. The new recipient receives a copy of the message without affecting the original.

Get Mail

The top line on your screen alerts you to the arrival of a message for you in the system mailbox; it displays the word *MAIL* at the top left corner of your screen.

From a Mail file, press **[ESC]** and then **[4]** to retrieve your mail from the system mailbox. The Get Mail function moves the message from the system mailbox to the mailbox you are using.

When you open *dmbox.msg*, your default mailbox, DeskMate automatically retrieves any mail addressed to you from the system mailbox. Use the Get Mail function only if mail arrives while you are using *dmbox.msg* or if you want to receive mail in a file other than *dmbox.msg*.

Option

Press **[ESC]** and then **[5]** to set or revise the values on the Mail Options screen, shown in Figure 8.2. The values on your Mail Options screen pertain to any mail that you create using the Mail application. Other Mail users can have different values within their directories.

<ESC> = Commands		Mail	6/29/85 9:00 am
MAIL OPTIONS			
Default Editor command:.....		/appl/desk/dmtext	
Save Messages Sent? (Y/N).....		N File:	
# of Minutes to check mail:.....		01	
DISTRIBUTION:	Use Return Receipt option? (Y/N).....	Y	
	Use Carbon Copy option? (Y/N).....	N	
	Use Blind Carbon Copy option? (Y/N).....	N	
SIGNATURE:	Use Signature file option? (Y/N).....	N	
NAME	DISTRIBUTION LIST		
super	fred mary sher smy vin		
project	fred greg samy duc gary		

Figure 8.2

You can change the settings on your Mail Options screen by entering different values for any of options (above the Distribution List). Press **[ESC]**, then **[1]** to use the Reset function, which restores any settings you have changed to the original values.

The first line indicates the text editor you want to use when you compose a message (using the Edit function). The system uses the editor provided with the Text application unless you specify an alternative.

To change editors, type the execution command for the text editor you prefer to use. Of course, the editor that you substitute must be installed on your system to be accessible to the Mail application. Include in this command any parameters required to pass a data filename to the alternate text editor.

On the second line, **Save Messages Sent? (Y/N)**, type **Y** **[ENTER]** if you want to save a copy of each message that you send. If you elect to save the messages, the screen prompts you for a filename to collect them. Mail adds the extension **.doc** to the filename (if you do not) so that it will appear as a Text file on your Main Menu. Type **N** **[ENTER]** to indicate that you do not want to save copies of your messages.

Note: If you open your mail storage file through Text, you can see each message that you sent. After you exit the Text file, Mail continues to save messages to the file, but you will be unable to display subsequent messages through Text. Use XENIX commands (**more** or **cat**) to display the complete file.

The next setting determines the interval, in minutes, between routine checks of the system mailbox. If there is mail for you in the system mailbox when Mail checks it, the word MAIL appears in the top left corner of your screen to alert you to the fact. The message appears on the top line of any DeskMate application. Type a number from 1 through 60 to set the frequency of these mail checks.

Distribution. The next set of values pertains to the distribution of mail that you send. On the Mail Option screen, you indicate only that you do or do not want a given feature to appear in the header portion of the Create Mail screen. Specify the users you want to receive the receipts or copies on the Create Mail screen. See the instructions for the Create function for more details.

Using the arrow keys, move the cursor to a setting you want to change. Type **Y** to include the option on the Create Mail screen or **N** to withhold it.

The Return Receipt option generates a message that acknowledges delivery of your message at its destination. Mail sends the receipt to the user you specify (on the Create Mail screen), typically yourself. The receipt distinguishes between delivery of the message to the recipient and actual display of the content of the message by the recipient. Return receipt messages appear in your mailbox as separate pieces of mail.

The Carbon Copy option lets you indicate which users are to receive a copy of a given message. The third option, dealing with blind carbon copies, lets you designate a second group of users who receive carbon copies; users in this group, however, are not mentioned as recipients on the message as sent.

Signature. The next setting on the Mail Option screen allows you to conclude each message that you compose with a standard signature block. Type **Y** to use the signature block, and then use the 4-line area immediately below the option line to indicate the signature you desire. The block can contain blank lines as well as text.

To omit the signature, type N **[ENTER]**. Omitting the signature does not erase the signature block you enter on this screen; it merely prevents its being routinely attached to mail you create.

Distribution List. Press **[ESC]** and then **[↓]** to move the cursor to the lower portion of the screen, the Distribution List. Pressing **[ESC]** and then **[↑]** returns you to the upper portion.

On the Distribution List, you can specify an alternate name, or alias, for a user or group of users. An alias consists of, at most, 10 characters, all of which must be lowercase alphabetic characters. It is often useful to assign an alias to an individual user because user names are frequently cryptic and hard to remember. For example, you can supply the alias **cindy** for her user name, **cls**, to assist your memory.

Assigning a single name to a group of users is helpful if you frequently send mail (or carbon copies) to a large, fixed group of users. For example, Figure 8.2 shows a sample group named **super** that includes all of the supervisory people among the Mail users. You can distribute a memo to all of the supervisors by addressing it to **super**, rather than naming every member of this group.

Use the arrow keys to move the cursor within the Distribution List. Additional rows scroll onto the screen as you need them. To add a new alias, press **[ESC]** and then **[0]**. The list provides a blank line on which to enter the alias and then the user or users.

Type the alias in the first column, press **[ENTER]**. Do not type any blank spaces in the first column. Then, type user names in the second column, leaving a space between each one. The width of the column is the only limit on the number of users included in an alias. Press **[ENTER]** to move to the next line.

Add a user to an existing alias by positioning the marker in the righthand column and then typing the user's name. Arrow keys enable you to move the cursor without erasing or overstriking existing characters. To verify a user name, press **[ESC]**, then **[↑]** to return to the upper portion of the screen. Then, press **[ESC]** followed by **[3]** to use the Users function.

To remove an alias from your Distribution List, position the marker on the item and press **[ESC]** and then **[9]** to use the Delete function. Use **[CTRL]** **[X]** to make deletions from the list of user names in the righthand column or edit the lefthand column.

Press **[ESC]**, then **[3]** when you want to execute the Users function. The Users function displays a complete list of every user and alias recognized by the Mail system. Use the arrow keys to scroll the list until it displays the name you want to see.

Press **[CTRL]** **[C]** or your **[BREAK]** key when you are finished using the function. For a more detailed description, see the discussion of the Create function and the Create Mail screen.

Press **[ESC]** and then **[Q]** to quit the Mail Options screen and return to the mail list on the initial Mail screen.

Merge

The Merge function enables you to move mail from one mailbox, or file, to another. You can save a single message or a block of messages in another file (new or existing).

Using the arrow keys, highlight the first message that you want to merge. If you want to define a block of messages that includes this one, press **[ESC]** and then **[7]** to use the Select function (described in the next section).

Then, press **[ESC]**, followed by **[6]** to use the Merge function. At the bottom of the screen, the prompt appears:

Merge to :

Specify a destination for the message or messages by entering a filename at the prompt. The Merge function compares the filename with existing files in the current directory and creates the file if it does not already exist. To specify a mailbox (new or existing) in another directory, enter the pathname as the destination.

When the merger is complete, the bottom lines of the Mail screen reappear, and the Select function is cancelled.

Select

Use the Select function to define a block of messages that you can then manipulate with other functions. After you define a block of messages, you can forward, merge, copy, or delete the messages as a unit.

Move the marker in the mail list to a message that you want to select. Press **[ESC]** and then **[7]** to select the message under the marker. Using the arrow keys, you can adjust the size of the

marked area to include messages above or below the initial selection.

When the highlighted area includes all the messages you want to consider in the block, choose another function to use with the selected messages. Execution of a subsequent command cancels the Select function. You can cancel the Select function at a prior point by pressing **[ESC]**, followed by **[7]**.

Copy

The Copy function lets you store a copy of a message or block of messages in a Text (ASCII) file. The file can occupy the current directory or, if you specify the complete pathname, a different directory.

To copy a single message, position the marker on the message in the mail list, and press **[ESC]**, then **[8]**. Copy a group of messages by first defining a block of messages with the Select function and then pressing **[ESC]**, followed by **[8]** to use the Copy function.

When you select the Copy function, the following prompt appears at the bottom of the screen:

Copy to :

Enter the filename or pathname for the destination file. Remember that the Copy function overwrites the contents of an existing file when it copies the new message to it. Use the Merge function to append (without overwriting) the message or block of messages to a Text (ASCII) file.

Delete

The Delete function deletes a single message or a group of messages. Position the marker on the first listing that you want to delete from the mail list.

To delete only the message under the marker, press **[ESC]** and then **[9]** to execute the Delete function. The item immediately disappears from the list.

To delete a series of messages, use the Select function to define a block of messages, and then press **[ESC]**, then **[9]** to delete them all.

Create

Press **[ESC]** and then **[0]** to create a message—either a brief message or a letter many pages in length. The Create Mail screen appears. (The Create Mail screen also appears when you press **[ESC]** and then **[2]** to use the Reply function.) It contains a heading in the upper part of the screen and an area for composing your brief message in the lower part.

Completing the Heading. The heading contains useful information for the identification and distribution of the message. The exact contents of the heading depend on the options you select on the Mail Options screen. If you enter the Create Mail screen using the Reply function, a number of the fields in the heading already contain data.

To cancel the contents of the Create Mail screen and start over, press **[ESC]**, then **[9]** to execute the Clear function.

The first line identifies the destination of the message. Type the login name or an alias for the destined recipient. For convenience, you can use the Users function to display a list of all mail users (including special distribution names defined through the Mail Options screen).

Press **[ESC]** and then **[3]** to display the list of mail users. A box is displayed in the lower portion of the screen that contains the existing user names you can enter in the current field.

Use the arrow keys to scroll through the list of users until you find a user name that you want to duplicate. Then, press **[ENTER]** to select the name under the marker. The name appears immediately on the current line of the heading (except the Subject line).

Continue to select names in this manner for the current field. Press **[ESC]** and then **[Q]** to eliminate the User List and resume direct entry on the current screen.

On the second line, identify the subject of the message with a brief description. This description appears both in the heading and in the mail list to identify the message.

As many as 3 additional items follow, depending upon the options you elect to use in the Mail Options screen. For each of these items, you can enter names of users you want to involve in the distribution of the message. Either type the appropriate user names or execute the Users function to identify these recipients. Press **[ENTER]** to move to the next item.

The first of these optional prompts lets you specify users to whom you want a return receipt delivered. Typically, you specify yourself in this blank. The return messages alert you to the status of the message. They tell you when the message has, in fact, been examined by the recipient.

In response to the second prompt, `Carbon Copy`, enter the names of those users to whom you want to send carbon copies of the message. This information is provided to the receiver of the message itself.

If you are using the Reply function to create a response to a message, the Mail system assumes that you want to send carbon copies of your response to the same users who received carbon copies of the original message. You can, of course, alter this information if you choose.

The final option, labeled `Blind Carbon Copy`, enables you to send carbon copies to users who are not identified to the receiver of the message. This is comparable to photocopying a message and placing it on someone's desk.

Typing the Message. After the information in the heading is complete, the cursor moves to the message portion of the screen. Type a brief message or press `[ESC]` and then `[2]` to use the Edit function. The Edit function lets you type a message using all the features of your text editor.

If you use the message area below the header, type a message on the lines provided. Use `[BACKSPACE]` to make corrections as you type. When you reach the right margin of the message area, words wrap to the next line.

If you select the Edit function to compose your message, the screen displays a page using your text editor. This editor is the editor for the DeskMate Text application unless you substituted another text editor using the Mail Options screen.

Included on the text page is the heading information for your message. Compose your letter using the functions and editing conventions of your text editor. Do not edit or reposition the header of your message while you are using the text editor. Instructions for using the Text application appear in Chapter 4, "Text."

When you finish your letter, press `[ESC]`, then `[Q]` to return to the Create Mail screen. If you are using an editor other than Text, type its exit command.

The Create Mail screen, when it returns, does not display the message portion of your newly created mail. To review or revise the message, use the Edit function.

Other Special Functions. Before you send your message, consider 2 other special functions on the Create Mail screen. Both appear in the menu of functions when you press **[ESC]**.

Attach Press **[ESC]** and then **[5]** to attach an existing document to the current message. A prompt appears at the bottom of the screen requesting the name of the text file to append to your message. You can attach only one file to a message. Type the filename and press **[ENTER]**.

Copy Press **[ESC]** and then **[8]** to save a copy of the entire message, including the information in the heading, as a Text (ASCII) file. At the bottom of the screen, a prompt appears requesting the name of the file into which you want to save the message. Type a filename and press **[ENTER]** to create the copy.

Sending A Message. When you finish the body of your message, press either **[ESC]**, then **[Q]** or **[ESC]** and then **[1]** to send it. Pressing **[ESC]** and then **[Q]** sends the message and returns you to the initial Mail screen. Pressing **[ESC]**, then **[1]** executes the Send function to send the message and redisplay the Create Mail screen so that you can compose another message.

Press **[ESC]**, then **[X]** to return to the initial Mail screen **without** sending the message. If you return to the Create Mail screen before you exit Mail, the unsent message is redisplayed.

Printing Mail

On the initial Mail screen, position the marker on a message or on the first of a series of messages that you want to print. To select a block of messages, press **[ESC]** and then **[7]**. Use the arrow keys to expand the marker until it includes all the messages you want printed.

Print the message or selected block of messages by pressing **[ESC]** and then **[P]**. The next screen displays the current printer settings. Alter any of these settings, and then press **[ESC]**, followed by **[Q]** to begin printing. When the printer stops, the Mail screen returns.

Exiting Mail

Press **ESC**, then **Q** to exit the Mail application and return to the previous level of activity—the Main Menu or your system prompt.

UTILITIES

The Utilities function expands the Main Menu to provide you access to 5 useful commands. These utilities let you make backups on floppy diskettes and restore the contents of backup files to your system. Other commands enable you to monitor current activity and to determine the free space available on various disks.

Selecting and Exiting Utilities

Select the Utilities function on the Main Menu by pressing **[ESC]** and then **[5]**. The list of applications in the last column changes to display the available utilities, as shown in the following sample screen, Figure 9.1.

<ESC> = Commands			Menu		11/21/85 10:30am
Nov 1985 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30			Events for Today: Bids due for Goren contract 10:00a Meet West to discuss convention 01:30p Appt. with Ruff at club Current Directory: /usr/sample/		
Text	Worksheet	Filer	Calendar	Mail	UTILITIES
Letters.doc Lhead.doc Labels.doc	Budget.wks	Clients.fil Vendors.fil	dmevents.cal Projects.cal	dmbox.msg Johnson.msg	BACKUP RESTORE FREE WHO ACTIVITY

Figure 9.1

To choose a utility, use the arrow keys to position the marker on the utility you want to use. Then, press **ENTER**. A window appears within the menu screen displaying the information and, in many cases, requesting further data.

Most of the utilities return you to the Main Menu after you execute them. Under other circumstances—if the utility does not return you to the Main Menu or if you want to return without executing the utility—press **ESC** and then **Q** or **CTRL C**.

Backup

The Backup utility enables you to save data from your files onto floppy diskettes. Backing up the contents of your files is good protection in the event that you accidentally delete files or lose data because of an equipment failure or power outage. Use the Restore utility, described in the next section, to copy the contents of your backup diskettes onto your operating disk once again.

When you select the Backup utility, the following prompts appear on the screen to let you specify the files or directories you want to save:

Backup

File or Directory:

Floppy Drive:.....0

The first prompt requests the name of the file or directory you want to back up. Press **ENTER** to back up the entire directory or enter the name of a file or directory in the current directory.

To back up a file or directory not in the current directory, type its complete pathname. You can specify a number of files or directories or a combination of the two. Be certain, if you do, to leave a space between items.

At the second prompt, identify the floppy disk drive at which you intend to make the copy. The number you type to identify a floppy drive depends on the type of diskettes you are using for the copy. The numbers 0 and 1 refer to floppy drives 0 and 1 respectively using high-density diskettes. The numbers 2 and 3 refer to floppy drives 0 and 1 respectively using standard-density diskettes.

Insert a formatted diskette in floppy drive 0 or 1. If you are backing up files onto a high-density diskette, press **[ENTER]** to use drive 0 or type 1 **[ENTER]** to use drive 1. If you are backing up files onto standard-density diskettes, type 2 **[ENTER]** to indicate floppy drive 0 or 3 **[ENTER]** to indicate floppy drive 1.

If the backup diskette contains data, the Backup utility overwrites its contents. Do not use the Backup utility to copy a file that is larger than 1.2 megabytes onto a high-density diskette or a file larger than 360K onto a standard-density diskette. XENIX provides backup commands for such files.

If you need more than a single diskette to back up all the files and directories you requested, a prompt appears instructing you to insert another diskette. Remove the first diskette, and replace it with another formatted diskette. Press **[ENTER]** to continue the backup procedure.

The message remains on the screen after you insert another diskette and continue. If you need to insert yet another diskette, the cursor moves to the end of the message again.

When the Backup utility finishes backing up the items you requested, the Main Menu reappears.

Restore

The Restore utility enables you to copy information from a backup diskette onto your disk. To use the utility, you need a diskette created using the Backup utility and a floppy disk drive.

Change the working directory to the one into which you want to restore the data from the backup diskette. Then, choose the Restore utility; the following screen appears:

```

      Restore

Floppy Drive:.....0

```

Insert the backup diskette in a floppy disk drive. At the prompt, press **[ENTER]** if you are restoring files from a high-density diskette on floppy drive 0. To identify floppy drive 1 instead, type 1 **[ENTER]** over the displayed value. If you are restoring files from standard-density diskettes, type 2 **[ENTER]** to indicate floppy drive 0 or 3 **[ENTER]** to indicate floppy drive 1.

When the Restore utility completes its task, the Main Menu reappears.

Free

Select Free to display the amount of free space on a given disk. The following screen appears:

```

                                Free

Device:.....usr
K bytes free =

Press <ESC><Q> to Quit
```

Use the first prompt to identify the disk drive you want to check. The displayed value specifies hard disk drive 0. To display the number of bytes free on this drive, press **ENTER**.

Using the indicated format, you can specify a different device, either another partition of the hard disk or a mounted file system or an alternate drive.

The free space is displayed in kilobytes (K). Each kilobyte equals 1024 bytes.

After the utility displays the free space on the specified drive, check another drive, or press **ESC**, then **Q** to return to the Main Menu.

Who

The Who utility enables you to determine which other users are logged onto the system. The utility displays a number of other items of information, arranged in successive columns. The following figure, containing sample data, illustrates the arrangement of information on the screen.

```

                                Who

Login      Name      TTY  Idle      When      Office
edwin      Edwin Raymond  01           Wed 13:36   2916
jill       Jill Roberts   03  14:47   Wed 12:31   2927

Press <ESC> <Q> to Quit
```

The first 2 columns reveal the user name and full name of each person using the system at that moment. The third column displays the terminal line number the person is using.

The next 2 columns, headed `Idle` and `When`, list the time at which the most recent activity occurred and the most recent login time for each user. Both columns give times in a 24 hour (military) format. The last column displays the office number or extension for each user.

Press `[ESC]` and then `[Q]` to exit the `Who` utility and return to the Main Menu.

Activity

`Activity` is a utility that displays a list of the processes that are running. When you select it, a screen appears such as the one in the following figure. The information `Activity` provides can help you if you are waiting to use a file that someone else is executing.

Activity					
User	Process	Start	TTY	Used	What
root	0	Dec 31	?	0:03	swapper
root	1	Dec 31	?	0:03	/etc/init
jill	144	14:59:20	01	0:01	-csh
edwin	1367	11:08:35	02	0:01	-sh -c ps -ef >uxtemp
root	1368	11:08:37	02	0:02	ps -ef
Press <ESC> <Q> to Quit					

The screen shows a series of data about each process. The first 2 columns identify the user who controls the process and the process itself. The screen specifies each process by a unique process identification number.

In the preceding figure, notice that a user called `edwin` controls the process with the identification number, 1367. The next column, headed `Start`, tells you the time of the day that the process began. The times appear in a 24-hour format that gives hours, minutes, and seconds. The fourth column indicates that Edwin began the process from teletype 02, the terminal connected to port 02.

Under the column headed `Used`, the utility displays the amount of processor time a given process has used in minutes and seconds. Edwin's process used 1 second or less.

The last column identifies the command that initiates a given process. Edwin's process follows the command sequence `-sh -c ps -ef >uxtemp`. A command sequence that goes beyond the right margin continues on the next line.

If Activity cannot display all the current processes on a single screen, it prompts you to press the space bar to display the rest of them. When you are finished examining the list, press `[ESC]`, then `[Q]` to return to the Main Menu.

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BEGINNING THE SAMPLE SESSION

During most of the sample session, you will be working with sample data involving Mr. Edwin Raymond's catering firm, Bon Appetit. You will be using DeskMate on Edwin's behalf.

Before you can begin running DeskMate, you must, of course, be operating under XENIX and you must install the program on your disk. Installing DeskMate is a simple procedure, described in Appendix A of the Reference section.

The sample session is divided into mini-sessions, each covering a different application. Although some benefit does derive from completing the chapters consecutively, they are largely independent of one another.

The sample session starts with an examination of the Main Menu in this chapter. Each subsequent chapter demonstrates a major application and one of the Desk Accessories. In this chapter, you will use the Messages accessory.

When you are finished with a particular part of the sample session and want to exit DeskMate, press `[ESC]` and then `[Q]` at the Main Menu.

Moving the Marker

On the screen, a *cursor* or a *selection marker* indicates your current position. The cursor is a single flashing character, while the selection marker highlights a block of characters.

In the sample session, you will use the arrow keys to move the cursor or position the marker on a particular item on the screen. To move the cursor or marker farther and faster, use `[ESC]` in combination with the arrow keys. See your reference manual for details on using the marker movement keys.

Executing Commands

The applications in DeskMate frequently use **CTRL** or **ESC** in combination with another key to execute a command. These *key sequences* are slightly different.

Use **CTRL** in much the same way you use the **SHIFT** key of a typewriter. When the manual shows **CTRL** preceding another key, press **CTRL** and hold it down while you press the second key.

To produce a key sequence using **ESC**, do not hold down **ESC** while you press the second key. After you press **ESC**, the command lines soon appear, listing general commands and functions specific to the current application or menu. Press a second key to execute the associated function.

Note that you do not have to wait for the command lines to appear if you are certain which combination key executes the command you want to use.

Differences in Terminals

Some terminals support keys that you can press instead of key sequences. Page keys can replace the use of **ESC** with the arrow keys, as shown in the following chart:

Standard keys

ESC	then	→
ESC	then	←
ESC	then	↑
ESC	then	↓

Optional key

PG RT
PG LT
PG UP
PG DN

You can also use function keys, such as **F1** or **F2**, to select functions. If you know which number corresponds to the function you want to execute, you can simply press the corresponding function key in place of **ESC** and the number. For example, press **F1** in place of **ESC** followed by **1**. (Press **F10** for **ESC** and then **0**.)

You can also press the function key instead of the number after **ESC**. Accordingly, **ESC** and then **F1** is equivalent to **ESC**, then **1** or **F1**.

The following figure shows other possible key substitutions. In each case, press the key in the right column in place of the key sequence in the left if your keyboard supports the alternate.

Standard key

CTRL	C
CTRL	W
CTRL	X
CTRL	T

Optional key

BREAK
INSERT
DELETE
HOME

Copying the Sample Files to Your Directory

To work through the sample session and have access to Edwin Raymond's files, execute a file that copies the sample data to your home directory. Actually, DeskMate places the files in a subdirectory of your home directory named `edwin/`.

Note: In the unlikely event that you have already created a subdirectory within your home directory named `edwin`, change its name or delete it before you use the sample session.

Complete the following series of steps to load and display the sample files:

1. Begin by logging on to the system. After you boot up under XENIX, the first prompt reads:

`login:`

Type your user name and press `ENTER`. The next prompt appears:

`Password:`

Enter your password. The characters you type do not appear on the screen. After you press `ENTER`, the system verifies the login information and displays your system prompt.

2. At the system prompt, type:

`dmsample` `ENTER`

This command executes a file that copies a sample directory within your home directory. You can delete it later.

Displaying the Main Menu

To display the files you copied, open DeskMate and then change the current directory. Follow the steps below:

1. Next, display the Main Menu for DeskMate. At the system prompt, type:

desk ENTER

DeskMate displays its Main Menu with your home directory as the current directory. Figure 10.1 illustrates the screen that appears:

<ESC> = Commands				Menu		11/20/85 10:30am					
NOV 1985				Events for Today:							
1 2 3											
4 5 6 7 8 9 10											
11 12 13 14 15 16 17											
18 19 20 21 22 23 24											
25 26 27 28 29 30											
				Current Directory: /usr/your home/							
Text		Worksheet		Filer		Calendar		Mail		APPLICATION	

Figure 10.1

The Main Menu displays a monthly calendar and a list of events scheduled (through the Calendar application) for today. Below the area for today's events is a line on which the menu highlights the current directory. On your screen, the current directory is your home directory. In Figure 10.1, *your home* represents the actual name of your home directory.

Across the middle of the screen, the Main Menu lists the DeskMate applications. If you have not yet used DeskMate, the lower part of the screen lists no files under the application names.

2. Change the current directory using the Directory function. Press **[ESC]** and then **[2]**. The following prompts appear at the bottom of the screen:

<ENTER> to Expand or
New Current Directory:

Type the name of the directory that contains the files for the sample session:

edwin **[ENTER]**

DeskMate redraws the Main Menu as shown in Figure 10.2.

MAIL <ESC> = Commands			Menu		11/20/85 1030am
Nov 1985 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30			Events for Today: Make appointment with accountant Mom's birthday - call florist Write confirmation letter to Wilson 05:30a Shop at fish and produce wholesale markets 07:30a Meet Bill at gym Current Directory: /usr/your home/edwin/		
Text	Worksheet	Filer	Calendar	Mail	APPLICATION
Address.doc Lhead.doc		Clients.fil	Events.cal	Mailbox.msg	

Figure 10.2

Figure 10.2 shows data in the upper right region of the menu, the area headed **Events for Today**. These events appear on your screen only if the current date corresponds to the date used in the sample session, November 20.

Notice the highlighted area below the list of today's events. In that space, the menu now identifies */usr/your home/edwin/* as the current directory. During the sample session, consider this directory to be Edwins's home directory. It contains the files that you will use as you proceed through the sample session.

The 6 columns that make up the lower part of the screen help you keep track of the files you use with your DeskMate applications.

The first 5 columns list files in the current directory grouped by application. The file `Address.doc`, for example, appears below `Text` because it is a document file that you read and edit through the `Text` application.

The sixth column, headed `APPLICATION`, lists applications (in addition to those in columns 1-5) that you can execute directly from your DeskMate menu. When you first install the DeskMate programs, this column is empty.

In the sample session, you open a number of files and then add to and manipulate the data in them. You will create and open files from the Main Menu as you complete the tasks in each chapter.

Using Messages Within the Main Menu

On the top line of your Main Menu, notice the word `MAIL`. The presence of this word on the top line indicates that the system is storing new mail for you. Check the mail from the Main Menu or from any application by executing `Messages`—one of the Desk Accessories.

1. Press `[ESC]` to display the label lines along the bottom of the screen. The label lines list command keys and the commands that they execute. Notice that `Desk Accessories` is paired with `<=>`.
2. Press `[=]` to display the menu of Desk Accessories. It appears as a window in the center of the screen, as shown in Figure 10.3.

MAIL <ESC> = Commands		Menu		11/20/85 1030am	
Nov 1985 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30		Events for Today: Make appointment with accountant Mom's birthday - call florist Write confirmation letter to Wilson 05:30a Shop at fish and produce wholesale markets 07:30a Meet Bill at gym Current Directory: /usr/your/home/edwin/			
Text	Worksheet	Desk Accessories	Mail	APPLICATION	
Address.doc Lhead.doc		Calculator Events/Alarm ** Messages Phone Directory Scratchpad XENIX Command <ESC> = Commands	dmbox.msg		

Figure 10.3

The stars to the left of Messages on the screen confirm the fact that there is new mail for Edwin.

- To examine the new mail, press **[M]** to use the Messages accessory. The Messages window replaces the menu of Desk Accessories. It identifies the new mail so that you can read the messages if you choose.

Figure 10.4 shows the Messages window. There are 3 new messages, the first from your own login name, the second from Dan, and the third from Jody. Edwin decides to look immediately at the message from dan.

From	Subject
you dan jody	Change of plans--Johnson wed Reassignment

<ESC> = Commands

Figure 10.4

4. With the marker highlighting the message from Dan, press  to display the contents of the letter. Dan's message soon appears.

Note: If you are a newly created XENIX user, you may have an additional message on your screen that welcomes you to XENIX (depending on your shell). Its header on the message screen identifies it by sender only (no Subject).

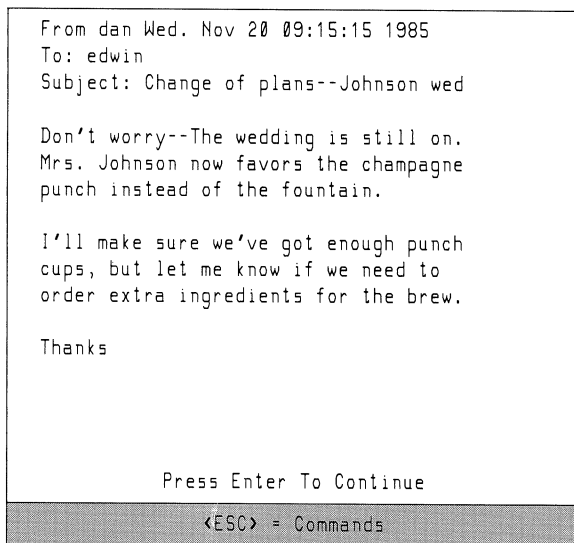


Figure 10.5

5. After you read the message, press **[ENTER]** to return to the list of new messages.
6. Press **[↓]** twice to move the marker to the last item, and press **[ENTER]** to read the message from Jody
7. After you read the message, press **[ENTER]** to return to the list of new messages.

Edwin decides to reply to his messages later in the day. In reading them, however, he is reminded that he needs to cancel the rental arrangements for the champagne fountain. He decides to send himself a note so that he will not forget this detail.

1. Press **[ESC]**, then **[0]** to create a message. The window changes to display a blank message area, headed by a series of prompts.
2. At the first prompt, **To:**, type your own user name, and then press **[ENTER]**.

3. At the second prompt, `Subject:`, type:

Johnson wedding reminder `ENTER`.

The cursor moves to the message area.

4. Type the following message. Words that overrun the end of the line wrap to the next line.

Don't forget to cancel the champagne fountain now that the Johnsons have opted for punch.

To correct errors, position the marker using `BACKSPACE` or the arrow keys. `BACKSPACE` erases characters, while the arrow keys do not affect them. Then, type over the incorrect portion of the message.

5. Press `ESC` and then `Q` to send the message. The list of messages returns after the message is sent.
6. To exit the Messages accessory, press `ESC` and then `Q`. Exiting an accessory returns you to the screen you left when you executed it. In this case, you chose the accessory from the Main Menu (which has remained in the background while you read and created messages).

When the screen displays the Main Menu once again, Edwin proceeds with the tasks outlined in the following chapters. Each chapter focuses on a different application. Edwin uses the accessories in the course of executing an application, just as you might in actual use of the DeskMate programs.

USING TEXT

In this chapter you will create a new text file and type a letter to one of Edwin's customers. As you do so, you merge data from other documents and use the editing features of the Text application. Finally, you print the letter.

To open a new Text file, follow the steps listed below. Use the arrow keys to position the marker.

1. Position the marker at the head of the first column (if it is not already there). It highlights the name of the application, `Text`.
2. Press `[ENTER]` to execute the application. The application prompts you for a filename at the bottom of the screen.

Enter filename:

3. At this prompt you can type the name of an existing file or invent a name for a new file. Type **Letter** `[ENTER]` for the name of the Text file you are creating.

Text opens a new Text file named `Letter.doc`. The application assigns the extension, `.doc`, if you do not include it. A blank typing page appears, the first page of your new document.

4. Press `[ESC]` to see the functions available to you in the Text application. They appear on the 2 bottom lines of the screen as shown in the figure that follows.

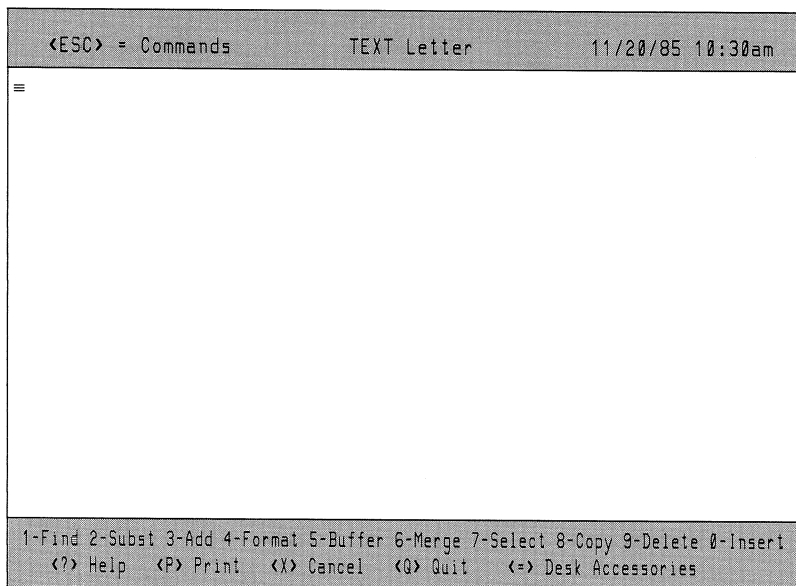


Figure 11.1

5. Press **[ESC]** to erase the function lines and display the full Text screen once again.

Entering Text

Edwir wants to send a letter to a client to confirm the dates of upcoming events. Begin by typing the following letter, pressing **[ENTER]** where indicated. Do not worry about typing mistakes; later you'll learn how to correct errors by inserting, deleting, and overstriking text.

Dear Mrs. Williams:

I am writing to confirm your agenda for the upcoming month. I have you scheduled for the following days:

12/3/85 Luncheon for 8 at noon

12/12/85 Afternoon tea for 6 at 3:00

12/25/85 Dinner for 10 at 8:00

12/31/85 Wedding reception for 60 at 7:30

If any of the above information is incorrect, please inform me as quickly as possible, as December appears to be a very busy month. Also, we need to get together soon and arrange the menus.

Sincerely,

Edwin Raymond

Raymond needs to add an event for December 1st to the list in his letter.

1. Move the cursor over the 1 of 12/3/85 by repeating until the cursor rests on the line containing the December 3rd event.
2. You are currently in Add mode (the default typing mode) as shown in the label line at the bottom of the screen. Type 12/1/85, press the space bar 3 times, and type **Bridal shower for 15 at 1:00** .

Note that in Add mode, the original text moved to the right as you inserted the new text and then moved down to the next line when you pressed after typing.

Next, change the information for the event scheduled on December 25th. The event now falls on December 26th for 12 people.

1. Move the cursor over the 5 of 25.
2. To change the typing mode to Replace, press , then . (Note that Rpl now appears in the label line when you press .)
3. Type 6 over the 5, move the cursor to the 0 of 10, and then type 2 over the 0. Replace (overstrike) mode lets you type over text.

You need to make one more correction—change the word `and` in the last sentence to `to`.

1. Move the cursor to the `a` in `and` and type `to`.
2. Press `[ESC]` and then `[9]` (or your deletion key) to erase the `d` and shift the rest of the sentence to the left one character.
3. Press `[ESC]`, then `[3]` to switch from Replace back to Add mode.

Note: If you made any mistakes while typing the letter, correct those errors now by using the Add/Replace typing modes and the Delete function, and then continue with the rest of the Text sample session.

Duplicating Text to Another File

Next, you want to insert address information from another Text file above the salutation.

1. Press `[ESC]`, then `[Q]` to save the letter and return to the Main Menu. When the Main Menu is redisplayed, notice that your new file, `Letter.doc`, appears in the column of Text files.
2. Move the marker to `Address.doc`, and press `[ENTER]`. The name and address information for Edwin's customers appears on the screen.
3. To look up Mrs. Williams's address, press `[ESC]` and then `[1]` to use the Find function.
4. Type `Wil` `[ENTER]` for the search string (using only part of the name, Williams). The cursor moves to the `W` of `Williams`.

Anytime you want to do something with a block of information (Buffer, Copy, Delete, Insert, Print), you must first define the text block using the Select function.

1. Move the cursor to the `M` of `Mrs.`, and press `[ESC]`, then `[7]` to select the beginning of the address block.
2. Press `[↓]` to select the 3 lines of Mrs. Williams's address block.

3. Press **[ESC]**, then **[8]** to save a copy of the address block as a new file. At the bottom of the screen, you see:

Enter copy filename:

Type **Williams** **[ENTER]** as the name for the new file.

4. Since you are finished using the Address file, press **[ESC]** and then **[Q]** to return to the Main Menu.
5. Next, return to the file containing the letter to Mrs. Williams. In the list of files, position the marker over `Letter.doc` and press **[ENTER]**.

The letter reappears with the cursor at the beginning of the letter (on the D in `Dear`).

6. To insert the address block above the salutation, press **[ESC]**, then **[6]** to use the Merge function.
7. At the prompt, type **Williams** **[ENTER]** to merge the contents of that file at the cursor.

You need to add a blank line between the customer's address and the salutation.

1. Press **[ESC]** to check the label line at the bottom of the screen. If `Rpl` is displayed instead of `Add`, press **[3]** to change typing modes. If `Add` appears, press **[ESC]** again.
2. Move the cursor to the D of `Dear`, and press **[ENTER]** to create a blank line.

Next, place Edwin Raymond's return address above Mrs. Williams's address. A Text file called `Lhead.doc` contains the standard heading Edwin uses at the top of all his correspondence.

1. Press **[CTRL]** **[T]** to move the cursor to the beginning of the text.
2. To insert the Text file, `Lhead.doc`, at the top of the letter, press **[ESC]** and then **[6]** for the Merge function; then, type **Lhead.doc** **[ENTER]** for the filename.
3. To add a blank line between the addresses, move the cursor to the M in `Mrs.` and press **[ENTER]**.

The address information appears at the beginning of the document, and now the letter looks very similar to this:

Edwin Raymond
4000 Seville Avenue
Fort Worth, Texas 76126
Date

Mrs. Eliot Williams
1908 Florida Avenue
Denton, Texas 70912

Dear Mrs. Williams:

I am writing to confirm your agenda for the upcoming month. I have you scheduled for the following days:

12/1/85 Bridal Shower for 15 at 1:00
12/3/85 Luncheon for 8 at noon
12/12/85 Afternoon tea for 6 at 3:00
12/26/85 Dinner for 12 at 8:00
12/31/85 Wedding Reception for 60 at 7:30

If any of the above information is incorrect, please inform me as quickly as possible, as December appears to be a very busy month. Also, we need to get together soon to arrange the menus.

Sincerely,

Edwin Raymond

4. Move the cursor to the D of Date in the letterhead block.
5. Press **[ESC]**, then **[3]** to switch to Replace mode; then type:

November 20, 1985

Note: With the addition of 2 address blocks, the letter now exceeds the maximum number of text lines that the screen can display. Press **[ESC]**, then **[↓]** to move the cursor to the bottom of the page. Repeat the key sequence to display the end of the letter to see the lines that do not fit on the screen.

Printing Text

Before printing text, it is a good idea to make the line width displayed on the screen coincide with the line width that you intend to print on a page. By using the function, Format, you can get a rough idea of what the text will look like when you print it.

1. To change the displayed line width, press **[ESC]** and then **[4]** for Format.
2. The preset value for the line width is 78. Since you plan to print the letter with a line width of 50, type **50** **[ENTER]** to change the display.

The letter now looks like this:

Edwin Raymond
4000 Seville Avenue
Fort Worth, Texas 76126
November 20, 1985

Mrs. Eliot Williams
1908 Florida Avenue
Denton, Texas 70912

Dear Mrs. Williams:

I am writing to confirm your agenda for the upcoming month. I have you scheduled for the following days:

12/1/85 Bridal shower for 15 at 1:00
12/3/85 Luncheon for 8 at noon
12/12/85 Afternoon tea for 6 at 3:00
12/26/85 Dinner for 12 at 8:00
12/31/85 Wedding reception for 60 at 7:30

If any of the above information is incorrect, please inform me as quickly as possible, as December appears to be a very busy month. Also, we need to get together soon to arrange the menus.

Sincerely,

Edwin Raymond

Note: Use the arrow keys to see the entire letter.

3. Be sure your printer is properly connected and on-line. Use standard 8 1/2 by 11 inch paper (80-column computer paper), and align the paper in the printer.
4. Press **[ESC]**, then **[P]** to display the printer settings. The screen shows the following settings.

```
Printer Settings

Lines per page:.....66
Printed lines per page:.....60
Left margin:.....0
Printed line width:.....80
Double spacing (Y/N):.....N
Print: Console, Local, Screen, File...C
```

5. The **Lines per page** refers to your paper size, the number of lines on the entire page. The displayed value, 66, applies to both regular-sized paper (8 1/2 by 11 inches) and wide, 132-column computer paper (14 by 11 inches). Press **[ENTER]** to use the displayed value of 66.
6. **Printed lines per page** refers to the number of lines you want printed on the page. This value equals the number entered for **Lines per page** minus the number of blank lines you want at the top and bottom of a page.

To compute the number of printed lines per page, first calculate the total for the top and bottom margins. When you align the paper to start printing, count the number of lines from the top edge of the paper down to the printer head position at which you want to begin printing. For equal top and bottom margins, double this number, and subtract the product from the total lines per page.

To have the top and bottom margins each contain 6 blank lines, subtract 12 from 66 and the result, 54, is the value for **Printed lines per page**. Type **54** **[ENTER]** to change this setting.

7. The displayed setting for the **Left margin** is 0. Type **15** **[ENTER]** to make the left margin approximately 1 1/2 inches from the edge of the paper.
8. The next setting, **Printed line width**, is the number of characters you want a printed line to contain. To change the preset line width of 80 to 50, type **50** **[ENTER]**.

9. To single space your printed document, press **ENTER** at the next prompt, `Double spacing`. This retains the displayed value, `N`.
10. At the last prompt, press **L** **ENTER** if you are connected to a printer at your terminal and your terminal supports local printing. If not, press **ENTER** to use the printer at the console. The printer begins to operate after you respond to the last prompt.

Substituting Text

1. After the printer stops, press **ESC**, then **Q** to save the document file and return to the Main Menu.
2. At the Main Menu, press **I** to move the cursor over `Address.doc` in the `Text` column, and press **ENTER**. The name/address information of Mr. Raymond's customers reappears on the screen.
3. Press **ESC** and then **?** to display the Text help screen. The screen shows:

```
----- Text Help -----  
  
To use a command, press <ESC>, then the number or letter:  
  
<ESC> <1>  FIND--Move cursor to first occurrence of search data.  
  
<ESC> <2>  SUBSTITUTE--Substitute replacement string for search  
           string. Press <Y> or <N> at each match, or  
           <CTRL C> to cancel search/replace process.  
  
<ESC> <3>  ADD/REPLACE--Change typing mode from Add (insert) to  
           Replace (overstrike). Repeat to switch back.  
  
<ESC> <4>  FORMAT--Change line width for document.  
           Minimum width is 21; maximum is 78.  
           <P>Go to previous page  <ENTER>Go to next page  <CTRL><C>Quit
```

4. To see the next help screen, press **ENTER**.

```
----- Text Help -----  
  
To use a command, press <ESC>, then the number or letter:  
  
<ESC> <5>  BUFFER--Copy selected block of text to copy buffer.  
<ESC> <6>  MERGE--Insert contents of another document at cursor.  
<ESC> <7>  SELECT--Define a block of characters to Save, Delete,  
           or place in Buffer.  
<ESC> <8>  COPY--Copy document or selected block in ASCII file.  
<ESC> <9>  DELETE--Delete character at cursor or selected block.  
  
<P>Go to previous page  <ENTER>Go to next page  <CTRL><C>Quit
```

5. Press to see more help screens. On the last help screen, returns you to your file and positions the cursor where you left it.

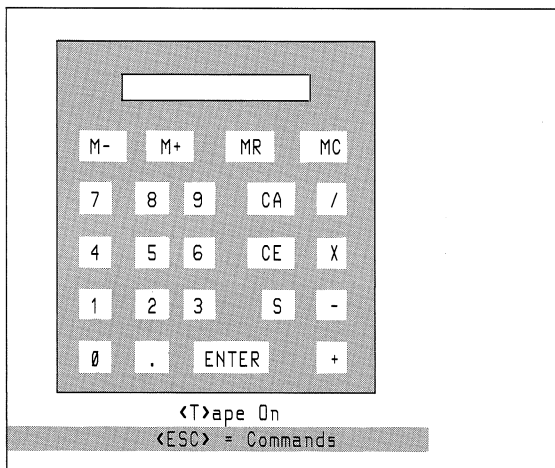
The only Text function you have not yet used is the Substitute function. This function lets you find a specific string of characters throughout a text file and replace the string with different text. When searching, the function ignores the distinction between upper and lower case. For example, `STRING` and `string` are recognized as equal.

1. Suppose you want to change almost every occurrence of Fort Worth to Dallas. Press and then for Substitute; then type **Fort Worth** at the prompt, Search string.
2. In response to the next prompt, Replacement string, type **Dallas** .
3. The cursor moves to the first occurrence of Fort Worth, which happens to be in Cindy Beauchamp's address, and you are asked Replace? (Y/N).
4. Press . The Substitute function replaces the search string, Fort Worth, with Dallas, the replacement string. Then, the cursor moves to the next occurrence of the search string, in Ellen McKinney's address.
5. Press to keep this occurrence of Fort Worth. The cursor moves to the last address, and a prompt appears, Replace? (Y/N).

6. Press **[Y]**. The screen displays the beginning of the document, indicating that there are no other occurrences of the search string.
7. To change the addresses back to their original states, press **[CTRL] [T]** to return to the top of the first page in the file. Press **[ESC]**, then **[2]** to select the Substitute function again.
8. This time, type **Dallas** **[ENTER]** for the search string and **Fort Worth** **[ENTER]** for the replacement string.
9. Press **[Y]** at the first occurrence, **[N]** at the second occurrence, and **[Y]** at the last occurrence. Now the Address file is the same as when you opened it.

Using Calculator Within Text

You can select the accessory, Calculator, while using any application. Before exiting Text and returning to the Main Menu, experiment with Calculator. To use the Calculator, first press **[ESC]**, then **[=]** to display the menu of Desk Accessories. When the menu appears, press **[C]** to select the Calculator accessory. The following screen is displayed, representing a common, hand-held calculator.



To perform an operation on the Calculator, press the keys on your keyboard that correspond to the Calculator keys. Where the Calculator key shows 2 characters, type both to produce an equivalent result. For example, type **ca** to produce the effect of pressing the Clear All key, **CA**, on the Calculator—to clear the current operation so that you can start over.

To use any of the commands from the command line, the highlighted strip below the calculator itself, press **[ESC]** followed by the key that executes the command. For example, press **[ESC]** and then **[3]** to create or edit Formula 3.

To practice using the Calculator, complete the steps that follow:

1. To add 5 and 1, type **5 + 1 [ENTER]**. The answer, 6, appears in the display window of the simulated calculator. Notice that this space shows, in turn, each operand, the operator (at the left), and then the result of the operation.

The operation also appears along the righthand margin of the screen. The column of figures represents the numbers that appear on a paper tape readout. The readout is a handy way to verify the numbers that you typed or results of previous operations.

2. To subtract 4 from 6 (the current entry), press **[=]** for subtraction; then, type **4 [ENTER]**. The entry line changes to 2.
3. To multiply the existing entry by 10.3, type the new operator, **x**, and then **10.3 [ENTER]**. The accumulator is now 20.6.
4. To divide 8.6 by .4, type **8.6/.4 [ENTER]**. The answer, 21.5, is shown on the entry line.
5. If you make a mistake in typing an operand, type **ce** before you press **[ENTER]** to perform the operation. When you use the CE (Clear Entry) feature, you erase only the current number. Previously entered operands and operators are unchanged.

For example, suppose you want to take 30% of the current value, and you leave out a decimal point. Press **x** for multiplication; then, type **30**. Type **ce** to erase the 30, and then type **.30 [ENTER]**. The new result is 6.45.

6. Typing **s** changes the sign of the operand from positive to negative and vice versa. For example, to divide 6.45 (the current entry) by a negative 4, press **/** for division. Then, type **4**, the divisor, followed by **s**, to change the sign of the operand to negative. Finally, press **ENTER**. The answer, shown in the display window (and on the tape readout), is -1.6125.
7. To exit Calculator and return to Text, press **ESC**, then **Q**. The Text screen you were using reappears.

To exit Text, press **ESC**, then **Q** to use the Quit function. The function saves the newly created document (or any editing changes made to an existing document) and returns you to the Main Menu.



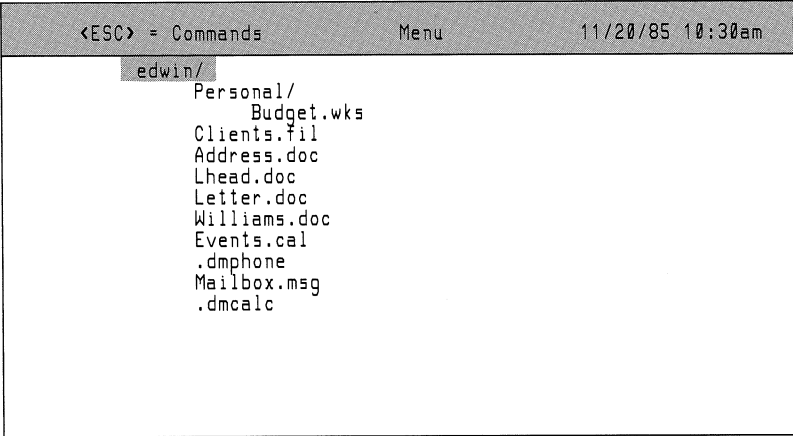
USING WORKSHEET

Edwin Raymond maintains a spreadsheet containing his home budget in a directory separate from his business files. Use the Directory function on the Main Menu to expand the directory, *personal/*, and then open the file from the expanded file list.

1. When the Main Menu appears, the current directory is the home directory, */usr/your home/edwin/* (for the sample session). Begin by executing the Directory function. Press **[ESC]** and then **[2]**. A prompt appears:

<ENTER> to Expand or
New Current Directory:

2. Press **[ENTER]** to expand the current directory. The screen displays a complete list of the files and directories within the current directory, *edwin/*.
3. Press **[↓]** until the marker highlights the directory you want to expand, *Personal/*, and press **[ENTER]**. The screen displays the files and directories within *Personal/*. Figure 12.1 shows the sample files as they appear on your menu screen.



```
<ESC> = Commands           Menu           11/20/85 10:30am
edwin/
  Personal/
    Budget.wks
  Clients.fil
  Address.doc
  Lhead.doc
  Letter.doc
  Williams.doc
  Events.cal
  .dmphone
  Mailbox.msg
  .dmcalc
```

Figure 12.1

4. Position the marker on *Budget.wks* in the newly displayed column of files. Then, press **[ENTER]** to open the file. The screen illustrated in Figure 12.2 appears.

<ESC> = Commands WORKSHEET Budget 02/25/85 10:30am						
1	2	3	4	5	6	7
1	BUDGET FOR JAN 1985					
2						
3	EXPENSE	BUDGET	ACTUAL	NET		
4	CATEGORIES	AMOUNT	AMOUNT	AMOUNT		
5						
6	Car Paymnt	250.00	250.00	.00		
7	Car Gas	80.00	60.00	20.00		
8	Home Gas	50.00	87.13	-37.13		
9	Electric	75.00	39.89	35.11		
10	Water	25.00	17.25	7.75		
11	Phone	50.00	61.10	-11.10		
12	Rent	400.00	400.00	.00		
13	Insurance	65.00	65.00	.00		
14	Grocery	150.00	113.57	36.43		
15	Fun	100.00	165.00	-65.00		
16						
17	TOTALS	1245.00	1258.94	-13.94		
18						
19						
20						
Select Command:						
R 1 C 1						

Figure 12.2

The top line of the screen identifies the file and displays the date and time. The remainder of the screen displays just a small portion of a worksheet on which you can enter and manipulate data.

A worksheet can fill as many as 99 columns and 99 rows, while a single screen of a worksheet (called a *window*) consists of 20 rows and 7 columns. The highlighted rectangular box that currently occupies Row 1, Column 1 (Cell 1,1) is called the *entry marker*.

The line near the bottom of the screen that now reads *Select Command* is the *command line*. It displays prompts and instructions to help you use the worksheet.

The blank space opposite *Select Command* is the *data entry area*. As you type data, it appears both on the data entry line and in the cell that the entry marker occupies. When you press **[ENTER]**, the data is *entered* into the cell, and the data entry line

displays the contents (if any) of the new cell. Pressing one of the arrow keys also enters data into a cell.

Below the command line is the *cell status line*. It identifies the cell currently occupied by the entry marker, the contents of the cell. In Figure 12.2, the cell status line shows R 1 C 1 (Row 1, Column 1) as the current cell. The cell is empty.

Creating a Simple Budget Worksheet

Edwin's budget compares budgeted amounts for expense categories with the actual amounts he spent during the month of January. In this chapter, you are going to re-create Edwin's worksheet, step by step, to learn how to use the Worksheet's basic functions.

1. Press **[ESC]**, then **[Q]** to return to the Main Menu. When the Menu reappears, set as it was when you left it, you see the expanded file system you displayed before opening the file, budget.wks.
2. Press **[ESC]** and then **[X]** to redisplay the entire Menu screen.
3. Position the marker on **Worksheet**, and press **[ENTER]**. Enter a filename for the spreadsheet you are about to create. Type **Example** **[ENTER]** for the filename. DeskMate creates and then opens a new file in the Worksheet application.

Entering Labels and Text Data

You begin by entering the column and row headings.

1. Press **[CAPS]** once to enter these labels in capital letters.
2. In Cell 1,1, the cell that the marker occupies initially type the heading, **EXPENSE**. Note that Enter text replaces Select Command to show the type of data the cell contains (text as opposed to numerical data).
3. Press **[↓]** to move the marker to Cell 2,1 (Row 2, Column 1), and type **CATEGORIES**.

Note: Remember that if you make mistakes in typing, pressing **[BACKSPACE]** deletes the previous character. Notice that pressing **[ENTER]** is unnecessary; moving the marker to a new location using an arrow key is sufficient to enter the data you typed.

4. Press **[→]** and **[↑]** to move the marker to Cell 1,2. Type **BUDGET**.
5. Press **[↓]** once to enter the data and change cells. Type **AMOUNT** to complete the Column 2 heading. (Don't be concerned at this point about the position of the column labels.)
6. Move the marker to Cell 1,3, and type **ACTUAL**.
7. Press **[↓]**. In Cell 2,3, type **AMOUNT** to finish the Column 3 heading.
8. In Cell 2,4, type **AMOUNT**.
9. Move the marker to Cell 1,4, and type **NET**.

To adjust the position of the labels, select a block of text and reformat its contents.

1. Move the marker to Cell 1,1. Press **[ESC]** and then **[7]** to use the Select function.
2. Select a block of cells by moving the marker to Cell 2,4.
3. To reformat the contents of all the highlighted cells, press **[ESC]** and then **[4]**.
4. A prompt appears on the command line that offers a series of formats:

Specify format types (L, R, D, I, \$)

At the cursor, type **R** **[ENTER]** to right justify the data in each of the selected cells. Cell contents shift to the locations shown in Figure 12.2.

Now enter the various expense categories in Column 1.

1. Press **[CAPS]** so that you can once again type both upper- and lowercase letters.
2. Move the marker to Cell 4,1 to enter the first expense category. Type **Car Paymnt**.
3. Press **[↓]** to move the marker to Cell 5,1; then type **Car Gas**.

4. Type the rest of the expense categories in Column 1.

Cell	Type
R6C1	Home Gas
R7C1	Electric
R8C1	Water
R9C1	Phone
R10C1	Rent
R11C1	Insurance
R12C1	Grocery
R13C1	Fun

5. Move the marker to Cell 15,1 to enter a label for Row 15. Press **[CAPS]**; then type **TOTALS**.

Figure 12.3 depicts the budget headings as they appear on your screen.

<ESC> = Commands WORKSHEET Example 02/25/85 10:30am						
1	2	3	4	5	6	7
EXPENSE	BUDGET	ACTUAL	NET			
CATEGORIES	AMOUNT	AMOUNT	AMOUNT			
3						
4 Car Paymnt						
5 Car Gas						
6 Home Gas						
7 Electric						
8 Water						
9 Phone						
10 Rent						
11 Insurance						
12 Grocery						
13 Fun						
14						
15 TOTALS						
16						
17						
18						
19						
20						
Enter text			TOTALS			
R15 C 1						

Figure 12.3

Entering Numbers and Formulas

With the headings in place, you can enter the budget figures:

1. To enter the budget amount for the first expense category, Car Payment, move the marker to Cell 4,2 and type **250** **[ENTER]** for the \$250.00 car payment. Note that **Select** command was replaced by **Enter number** when you typed the data, reflecting a numeric entry in the cell.

Since the built-in display format for numbers is for financial data (the dollar format) with 2 decimal places, .00 was automatically added to the 250 you entered. Also, note that a number is right-justified within a cell, whereas text is left-justified.

2. Press **[↓]** to move the marker to Cell 5,2; then type **80** to enter the budget amount of \$80.00 for gasoline.
3. Press **[↓]** and type **50** for the budget amount for Home Gas.
4. Type the budget amounts for the rest of the expense categories in Column 2.

Cell	Type
R7C2	75
R8C2	25
R9C2	50
R10C2	400
R11C2	65
R12C2	150
R13C2	100

Next, enter a formula to add these numbers and come up with a figure for the total budget.

1. Move the marker to Cell 15,2, and press **[ESC]** and then **[3]** to use the Formula function.
2. Type **SUM(R4)** **[ENTER]**. This formula tells Worksheet to add the numbers starting from Row 4 to Row 14, the row above the current location of the entry marker. This is a simple way of entering the formula:

$R4 + R5 + R6 + R7 + R8 + R9 + R10 + R11 + R12 + R13 + R14$

Note: Press **[CTRL]** **[F]** to erase values stored in formula cells.

3. Press **[ESC]**, then **[!]** to calculate the formula. The total budgeted amount, \$1,245.00, soon appears in Cell 15,2.
4. To enter the actual amount spent for the first expense category, Car Payment, move the marker to Cell 4,3, and type **250**.
5. Press **[↓]** to move the marker to Cell 5,3; then type **60** to enter \$60.00 for the amount actually spent for gasoline.
6. Type the actual amounts for the rest of the expense categories in Column 3.

Cell	Type
R 6 C 3	87.13
R 7 C 3	39.89
R 8 C 3	17.25
R 9 C 3	61.10
R10 C 3	400
R11 C 3	65
R12 C 3	113.57
R13 C 3	165

Now, enter a formula for calculating the total amount actually spent.

1. Move the marker to Cell 15,3 and press **[ESC]**, then **[3]**.
2. Type **SUM(R4)** **[ENTER]**. The formula computes the sum of the entries from Row 4 to the row containing the formula.
3. Press **[ESC]** and then **[!]** to calculate the formula. The total amount actually spent, \$1,258.94, soon appears in Cell 15,3.

You need to enter another formula to calculate the net amounts in Column 4.

1. Move the entry marker to Cell 4,4, and press **[ESC]**, then **[7]** for Select.
2. Press **[↓]** 11 times to indicate that Rows 4 through 15 are a single block because you want to calculate all values in Column 4 using the same formula.

3. Press **[ESC]**, then **[3]** and type **C2-C3** **[ENTER]**. This formula takes each budget amount in Column 2 and subtracts the corresponding actual amount in Column 3. It displays the net amount for each particular expense category in Column 4.
4. Now, press **[ESC]** and then **[I]** to calculate the net amounts. The computed results are displayed row by row, expense category after expense category.

Enter a title for the worksheet as a finishing touch.

1. Press **[CTRL]** **[T]** to move the cursor to Cell 1,1.
2. To insert 2 blank rows to make room for the title at the top of the worksheet, press **[ESC]**, then **[←]** to move the marker to the column containing the row number labels, and then press **[ESC]**, followed by **[0]** twice to insert 2 rows.
3. Press **[→]** twice to move the marker to Cell 1,2.
4. Type **BUDGET FOR**.
5. Press **[→]**; then press the space bar once, and type **JAN 1985** **[ENTER]**.

Printing a Worksheet

Now that you have finished creating the budget worksheet, you are ready to print it. Before using the print function, be sure that your printer is on line and that the paper is advanced so that printing will begin about an inch from the top of the paper (about 6 lines from the top).

1. Press **[ESC]** and then **[P]** to print; the screen displays the printer settings.
2. Press **[ENTER]** or **[↓]** to accept each of the settings. At the last setting, type **L** **[ENTER]** at a terminal to use your printer if your terminal supports local printing. Otherwise, press **[ENTER]** to use the console printer instead.
3. The printer begins to operate after you answer the last prompt. The printed copy looks like the report in Figure 12.4.

BUDGET FOR JAN 1985

EXPENSE CATEGORIES	BUDGET AMOUNT	ACTUAL AMOUNT	NET AMOUNT
Car Paymnt	250.00	250.00	0.00
Car Gas	80.00	60.00	20.00
Home Gas	50.00	87.13	-37.13
Electric	75.00	39.89	35.11
Water	25.00	17.25	7.75
Phone	50.00	61.10	-11.10
Rent	400.00	400.00	0.00
Insurance	65.00	65.00	0.00
Grocery	150.00	113.57	36.43
Fun	100.00	165.00	-65.00
TOTALS	1245.00	1258.94	-13.94

Figure 12.4

- After the printer stops, press **[ESC]**, then **[Q]** to save the worksheet and return to the Main Menu.

Setting Up an Amortization Table

Next, you may want to experiment with some of the Worksheet's more complicated and sophisticated features. In the following example, you create a spreadsheet for an amortization schedule.

For each period, the table calculates the fixed monthly payment and breaks it down into its 2 components: the interest and principal payments. There are 3 variables in this example: the original amount of the loan, the interest rate, and the number of periods over which you want to amortize the loan. The spreadsheet has 2 parts; set up the top part for entering values for the variables and the bottom part for the actual amortization table.

- On the Main Menu, position the marker on the Worksheet application. Press **[ENTER]** to execute the application. Because you did not indicate a particular filename, the bottom line prompts:

Enter filename:

2. Type **Table** for the filename of the worksheet you are creating, and then press **ENTER**.
3. As you begin, the marker is at the top of the first column. At Cell 1,1, type **LOAN AMT**.
4. Press **↓** to move the entry marker to Cell 2,1 (Row 2, Column 1); then type **INT RATE**.
5. Next, move the entry marker to Cell 3,1, and type **PERIODS**.

When you use the Calculate function, you want the function to ask for these 3 values. To prompt these requests, enter the following formulas in the cells to the right of the labels:

1. Move the entry marker to Cell 1,2, next to **LOAN AMT**, press **ESC** and then **3** for Formula, and type **?LOAN** **ENTER**. The word following ? tells you which value the worksheet needs for its calculation.
2. Move the entry marker to Cell 2,2, press **ESC**, then **3**, and type **?INTEREST** **ENTER**, so that later you will be prompted to enter a constant value for the interest rate.
3. Move the entry marker to Cell 3,2, press **ESC**, then **3**, and type **?PERIODS** **ENTER**.

The next step is to enter headings for the 7 columns.

1. Move the entry marker to Cell 5,1, and type **Period**.
2. Press **→**, and type **Balance**.
3. Press **→** to move the cursor to Cell 5,3, and type **Payment**.
4. In Cell 5,4, type **Interest**, and press **↓**; then right below *Interest* in Cell 6,4, type **Payment** to complete the heading for Column 4.
5. In Cell 5,5, type **Principal**, press **↓**, and type **Payment**.
6. Move the entry marker to Cell 5,6, and type **Cumulative**; then in Cell 6,6, type **Interest** to complete the Column 6 heading.
7. For Column 7, the last heading, type **Cumulative** in Cell 5,7, and type **Principal** in Cell 6,7.

Before you enter formulas, format the spreadsheet so that Columns 6 and 7 don't run together. To create wider columns, change the present width of all columns from 10 to 11 spaces.

1. Press **[ESC]** and then **[↑]** to move the entry marker to the top row; then press **[ESC]**, then **[↑]** a second time to move the entry marker to the line containing the column numbers.
2. Press **[ESC]**, then **[4]** to use the Format function, and type **ALL,11** **[ENTER]** to change the width of all columns to 11 characters.

Entering the Amortization Formulas

The next step is to enter formulas for these 7 columns. Column 1 lists all periods the loan covers. For example, this is a 1-year loan, and thus has 12 periods.

1. Move the entry marker to Cell 8,1, and type **1** **[ENTER]**.
2. Move the entry marker to Cell 9,1, and press **[ESC]** and then **[7]** for Select.
3. Press **[↓]** 10 times to indicate that Rows 9 through 19 are a single block and that all values in Column 1 are calculated using the same formula.
4. Press **[ESC]**, then **[3]** for Formula, and type:

R8+1 **[ENTER]**

The formula tells Worksheet to take the value in the preceding row, add 1, and display that value in the current cell. The cell labels in the formula adjust for each successive cell it occupies.

For example, move the entry marker to Cell 19,1. The value in the last row selected, Row 19, will equal the value in Row 18 (11) plus 1, or 12, the last period in the loan. The original formula, **R8+1**, has changed for each row so that when the value in Row 19 is calculated, the formula is **R18+1**.

5. Press **[ESC]**, then **[!]** to calculate. The numbers that appear identify each of the 12 periods, but they appear in the standard number format, the dollar format. Worksheet displays each period number with 2 decimal places and justifies each one on the right side of the column.

Next, you are going to format Column 1 so that the period numbers are whole numbers and so that they don't run into the calculations that will appear in Column 2.

1. Move the entry marker to Cell 8,1, press **[ESC]** and then **[7]**, and then select Rows 8-19.
2. Press **[ESC]**, then **[4]** for Format, and type **LI** **[ENTER]**.

L stands for left-justified, which means that the contents of all the selected cells will be flush left within the cell instead of the (default) right-justified format for numbers and calculated values.

You also specified an integer **(I)** format, since the period numbers do not need to be shown in dollar format (2 decimal places).

Column 2 shows the balance, the unpaid principal portion of the original loan amount. In the beginning, the balance is the full amount of the loan. The balance decreases steadily as each payment comes due.

1. Move the entry marker to Cell 8,2. The balance for Period 1 is the entire amount of the loan. When you calculate the values in the table, you enter a number to replace the variable in Cell 1,2.
2. Press **[ESC]**, then **[3]** to enter a formula, and type **R1C2** **[ENTER]**. This formula simply makes the value in Cell 8,2 equal to that in Cell 1,2.
3. Move the entry marker to Cell 9,2, and press **[ESC]** and then **[7]** for Select; then use the arrow keys to highlight Rows 9-19.
4. Press **[ESC]**, then **[3]** and type:

R8C2-R8C5 **[ENTER]**

This formula takes the value in the preceding row and the same column (the balance of the previous period), subtracts the value in the preceding row in Column 5 (the principal payment of the previous period) and displays the result.

For example, the value in the last row selected (the balance of Period 12), Row 19, equals the Period 11 balance in Row 18 minus the principal payment paid in Period 11, shown in cell 18,5. When the value in Cell 19,2 is calculated, the

original formula becomes to R18C2 - R18C5. (You can move the entry marker to Cell 19,2 to see that this is true.)

All values in Column 3 are the same to show the fixed payment that is paid every month on the loan.

1. Move the entry marker to Cell 8,3, and press **[ESC]**, then **[7]**; then select Rows 8-19.
2. Press **[ESC]** and then **[3]**, and type:

#R1C2*#R2C2/(1-1/(1+#R2C2)!#R3C2) **[ENTER]**.

The number sign (#) preceding a cell number indicates that the cell location is absolute, not relative to the cell in which the formula is entered. In other words, the original formula entered does **not** change for each row.

Note: This formula written in normal fashion is:

Fixed payment = (Loan Amt. x Int. Rate)/(1 - 1/(1 + Int. Rate)ⁿ) where Loan Amt. = original amount of entire loan, Int. Rate = interest rate per period, and n = number of periods.

This complicated-looking formula defines the numerator as the value in Cell 1,2 (**LOAN AMT**) multiplied (*) by the value in Cell 2,2 (**INT RATE**). The denominator is 1 minus 1 divided by (/) the quantity, 1 plus the value in Cell 2,2, raised to the value in Cell 3,2. The number of **PERIODS** becomes an exponent indicated by !

Column 4 shows the interest portion of each payment, which is the balance for a period multiplied by the interest rate.

1. Move the entry marker to Cell 8,4, press **[ESC]**, then **[7]**, and select Rows 8-19.
2. Press **[ESC]**, then **[3]**, and type:

#R2C2*C2 **[ENTER]**

For a particular period, this formula takes the value in the same row in Column 2 (the balance for that period) and multiplies it by the interest rate you enter in Cell 2,2.

The Principal payment of each period, the part of the total payment that actually goes to paying off the balance of the loan, is displayed in Column 5.

1. Move the entry marker to Cell 8,5, and press **[ESC]** and then **[7]**; then select Rows 8-19.
2. Press **[ESC]**, then **[3]**, and type:

C3-C4 **[ENTER]**

This formula takes the total payment (the value in Column 3) and subtracts the interest payment for that period (the value in Column 4) to come up with the principal payment for the period.

Column 6 shows the cumulative interest, the interest paid-to-date for each period.

1. Move the entry marker to Cell 8,6, and press **[ESC]**, then **[7]**; then select Rows 8-19.
2. Press **[ESC]** and then **[3]**, and type:

CMT(#R8C4) **[ENTER]**

This formula gives the accumulated totals for all 12 periods plus the final total of all the values in Column 4, starting with Row 8, and displays these values in Column 6.

For example, the interest paid-to-date for Period 4 is displayed in Cell 11,6. It is the sum of the interest paid in each of the first 4 periods—the values of Cells 8,4, 9,4, 10,4, and 11,4. CMT stands for *column summation*, and #R8C4 tells Worksheet to start the cumulative summing from Cell 8,4 (the interest paid in Period 1).

The last column displays the cumulative principal, the principal paid-to-date for each period. (After all 12 periods have been calculated, the last figure in this column, the cumulative principal for Period 12, equals the original amount of the loan.)

1. Move the entry marker to Cell 8,7, press **[ESC]**, then **[7]**, and select Rows 8-19.
2. Press **[ESC]**, then **[3]**; then type:

CMT(#R8C5) **[ENTER]**

The increasing values in this column show how the loan is gradually being paid off and retired.

Calculating and Reformatting a Worksheet

The spreadsheet is now completely set up, and you are ready to perform calculations.

1. Press **CTRL** **T** to move the entry marker to Cell 1,1. This example involves an 18% 1-year loan for \$1000.00.
2. Press **ESC** and then **!** to calculate. You are prompted to enter the **LOAN AMT.**
3. Type **1000** **ENTER**.
4. Type **0.015** **ENTER** for the **INT RATE**. This value represents the annual interest rate (18% or .18) divided by 12 to yield the monthly interest rate (1.5% or .015).

Note: The built-in display format for numbers is the \$ format with 2 decimal places. Although you cannot see the 5 you entered for .015, it is in memory and figures in the calculations.

5. Type **12** **ENTER** for the number of **PERIODS**. After you enter a value for the last constant, the computed results appear row by row, period after period.

You need to make a few final touches to make the spreadsheet look more professional. Although you want to display the data in the amortization table in dollar and cents format (2 decimal places), you want the interest rate cell to include up to 4 decimal places to cover the most common interest rate possibilities.

Second, you want the number of periods to appear as a whole number (without decimal places). Last, you want to justify the column headings along the right side of the column.

1. Move the entry marker to Cell 2,2, and press **ESC**, then **4** for **Format**.
2. Type **D** **ENTER** so that you can change the default number of decimal positions; then type **4** **ENTER**. Now you can see the entire entry, 0.0150, rather than the approximated entry, 0.02.
3. To specify an integer format for the number of periods entered, move the entry marker to Cell 3,2, press **ESC**, then **4**, and type **I** **ENTER**.

4. Move the entry marker to Cell 5,2, and press **[ESC]** and then **[7]**.
5. Press **[ESC]**, then **[→]** to select all the columns in the current window through Column 7.
6. Press **[↓]** to select the label lines.
7. Press **[ESC]**, then **[4]** for Format; then type **R** **[ENTER]** to right-justify the contents of all selected cells.

Entering Free-Form Text

There are 2 ways you can enter text: by cell or by block.

- For simple row and column labels that don't require more than 1 or 2 cells, position the entry marker on the cell in which you want the text to appear; then type the text and press **[ENTER]** (as you did when entering the column and constant labels).
- To type a paragraph or block of text, use the Select function to define the area in which you want to type; then use the Text function to type the desired text.

The Text function lets you type “free-form” text rather than cell by cell. For example, suppose that you want to add an explanatory note to the amortization table.

1. Move the entry marker to Cell 21,1, and press **[ESC]** and then **[7]**.
2. Press **[↓]** once to include the next row; then press **[→]** 4 times.
3. Press **[ESC]**, then **[2]** for Text.
4. Type the following:

**NOTE: Personal loan received 2/28/85 from
Saginaw Credit Union.**

Notice that the Text function wraps words when you reach the right margin of your text block.

5. To exit the Text function, press **[ESC]**, then **[Q]**.

When you type in a text block, you have access to limited editing features, such as deleting, inserting, and formatting text.

See the Reference Material on “Worksheet” for details on editing text within the Worksheet application.

Printing a Large Worksheet

Before using the print function, be sure that your printer is on line and that the paper is advanced so that printing will begin about an inch or so from the top of the paper (about 6 lines from the top).

Since the amortization table is larger than a single window, you need to select the area you want to print and then use the Print command.

1. To move the entry marker quickly to Cell 1,1, press **[ESC]** and then **[1]** for Find. You can use the Find function to search for a specific string of characters (or numbers) or a specific cell.
2. Type **R1C1** **[ENTER]** for Cell 1,1.
3. Press **[ESC]**, then **[7]** for Select.
4. Press **[ESC]**, then **[→]** to select Columns 1-7.
5. Press **[ESC]** and then **[↓]**, then press **[↓]** twice to select Rows 1-22.
6. Be sure that your printer is ready; then press **[ESC]**, then **[P]**.

You need to make a few adjustments to the printer settings before you begin to print.

1. Type **0** **[ENTER]** for Left margin and **79** **[ENTER]** for Printed line width.
2. Type **L** **[ENTER]** if you are at a terminal that supports local printing. Otherwise, press **[ENTER]** to accept the displayed value and use the console printer.
3. The printer begins to operate after you respond to the final prompt.

Sample Session

Your printout should look like this. (Compare your figures to be sure you entered all formulas correctly.)

LOAN AMT 1000.00
INT RATE 0.0150
PERIODS 12

Period	Balance	Payment	Interest Payment	Principal Payment	Cumulative Interest	Cumulative Principal
1	1000.00	91.68	15.00	76.68	15.00	76.68
2	923.32	91.68	13.85	77.83	28.85	154.51
3	845.48	91.68	12.68	79.00	41.53	233.51
4	766.49	91.68	11.50	80.18	53.03	313.69
5	686.31	91.68	10.29	81.39	63.32	395.08
6	604.92	91.68	9.07	82.61	72.40	477.68
7	522.32	91.68	7.83	83.85	80.23	561.53
8	483.47	91.68	6.58	85.10	86.81	646.63
9	353.37	91.68	5.30	86.38	92.11	733.01
10	266.99	91.68	4.00	87.68	96.12	820.68
11	179.32	91.68	2.69	88.99	98.81	909.67
12	90.33	91.68	1.35	90.33	100.16	1000.00

Note: Personal loan received 2/28/85 from Saginaw
Credit Union.

Recalculating a Worksheet

One of the most useful features of your worksheet is its capacity to recalculate the payment information rapidly, based upon changes in the loan data. Each time you execute the Calculate function, you can enter new values for the loan amount, rate of interest, and number of periods. The columns of payment data change accordingly.

1. To see the powerful recalculation ability of the spreadsheet application, press **[ESC]**, then **[!]**.
2. Suppose that you are calculating a 1-year loan for \$1000 at 15%. Because the previous value was also 1000, press **[ENTER]** for the LOAN AMT in Cell 2,1.
3. For the new interest rate, type **0.0125** **[ENTER]** (15% annual interest divided by 12 months per year and expressed as a decimal).

4. For number of periods, press **[ENTER]**. The lengthy, detailed calculations are performed almost instantly, saving you a considerable amount of time and effort. Note that the lower interest rate means a decrease in the fixed payment, shown in Column 3, from \$91.68 to \$90.26.
5. When you are finished examining your worksheet, press **[ESC]** and then **[Q]** to save your file and return to the Main Menu.

Note: As in the Text application, there is a second way of exiting Worksheet. Press **[ESC]**, then **[X]** if you change the contents of a worksheet and then decide you prefer to keep the unedited version. **[ESC]**, then **[X]** lets you retain the original worksheet and exit the application.

